

Alpha's Tale

The Economic Value of CSR

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Gabriel A. Huppé

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Abstract

Numerous studies report empirical evidence that trading strategies based on CSR information – often going ‘long’ the CSR ‘leaders’, and ‘short’ the ‘laggards’ – produced risk-adjusted investment profits (CSR Alpha). The claim of CSR Alpha is often used by SRI proponents and investment managers to convince mainstream investors that they can do well by doing good, potentially offsetting opportunity costs incurred by ethics-/values-based screening.

Although the thesis is attractive, claiming the presence of CSR Alpha begs an important question: Did the high (low) social responsibility standards of companies cause the strong (weak) stock returns—or—were these returns caused by risk or other underlying – perhaps time-period-specific – factors that happen(ed) to be correlated with CSR performance? SRI proponents would like to claim causality between CSR and returns, but such a causality relationship implies that CSR Alpha was the product of stock market inefficiencies; investors would have misevaluated the importance of CSR performance for future operating cash flows.

Our findings suggest that CSR Alpha arose because investors have historically overlooked the full value relevance of CSR information, thus being surprised at the earnings performance of CSR leaders and re-evaluating their assessment of companies’ prospects upon earnings announcements. Using market-based measures such as earnings announcement surprises, analyst forecast errors and Tobin’s *q*, the paper lends credence to the claim that CSR caused stock returns. However, proposing that recent institutional pressures to incorporate ESG factors have improved investors’ attention to and hence treatment of CSR information (eg. UN PRI), I find evidence suggesting that the stock market now more fully reflects the value relevance of CSR information, laying doubt on the view that best-in-class strategies will continue to outperform.

These findings shed light on the future viability of responsible investment strategies, implying that focus will move away from security selection and towards new approaches to strategic asset allocation, impact investing, and shareholder engagement. These latter strategies will be increasingly recognized for their potential to generate economic value. As the paper suggests, CSR *creates* shareholder value. Disputing the anti-thesis of reverse causality that relies on slack resources theory, it brings clarity to a 40 year-old debate that questioned the economic value of CSR.

I discuss the implications of CSR’s suggested economic role, the capital market’s appreciation of this role, and explore how institutions can affect more economically efficient and sustainable behaviour at the level of capital markets and strategic management. Findings yield substantial support for the work of institutional players like the UN PRI for helping improve capital market efficiency and thus affecting more economically efficient and sustainable corporate behaviour. The tale of CSR’s Alpha borrows from a rich assortment of over 500 scholarly articles for its foundation. Applied to actual developments in the investment industry, it yields significant findings for academics and practitioners alike in the fields of sustainability, management, capital markets, investment, finance, political science, law, ethics and institutionalism.

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Key words

socially responsible investing (SRI), corporate social responsibility (CSR), market efficiency, externalities, behavioural finance, asset pricing, learning, attention, institutional theory, isomorphism, UN PRI, Wages of Social Responsibility (WSR)

I. Introduction

Many factors are known to affect future operating performance and thus the value of a company. First there are industry-level considerations such as barriers to entry, supplier power, customer power and competitive rivalry, and then there are more specific company-level, resource-based factors that contribute to a company's competitive position within its industry and to its ability to address the greater social, economic, legal, political and technological forces that shape its external environment. To take account of these factors, investors' processes are separated into top-down and bottom-up components to evaluate both, a company's present and future operating context, and its ability to navigate this environment in the near and more distant future.¹

Investors – whose market role is to put an economic value on these companies – have long incorporated most such company-level factors into their expectations of future operating performance and thus valuation. However, there exists evidence that some such factors have historically been overlooked by the market: environmental, social and governance (ESG) standards are some such factors that, despite strong and persistent empirical evidence of their positive relationship with operating performance and abnormal stock returns, have only recently come to the fore of investors' attention.

For example, with regards to governance performance, Gompers et al. (2003) ranked companies by corporate governance standards and documented that an investment strategy that was 'long' the top companies and 'short' the bottom would have exploited the corporate governance attention deficit and produced an annual investment profit of 8 percent p.a. excess risk-adjusted returns (Governance Alpha) in years 1990-2000. With regards to social and environmental performance Statman & Glushkov (2009) ranked companies by social and environmental standards – otherwise known as corporate social responsibility (CSR) standards – and find that an investment strategy that was 'long' the leaders and 'short' the laggards would have produced an annual 6.12 percent p.a. excess risk-adjusted return for the period 1992-2007. Using different CSR rankings and similar trading strategies, other works find the same evidence of significant risk-adjusted outperformance (Derwall et al., 2005; Edmans, 2010; Kempf & Osthoff, 2007).

Since these strategies were based on publicly available information, semi-strong market efficiency (Fama, 1970) would require that these large and significant investment profits be the

¹ Managerial expertise, corporate reputation, corporate culture, compensation structures, R&D capabilities, advertising, other capital expenditures, relative market share and size, as well as the overall strength of the company's balance sheet, amount of leverage, solvency considerations, and exposure to other systematic risks are some of the most widely cited examples of such company-level, industry-relative factors that are shown to affect this ability.

product of underlying risk factors or the presence of period specific drivers.² Otherwise, these profits would be the product of market inefficiency; in which case investors are expected to eventually learn to properly incorporate ESG information into their investment processes (Lo, 2005). So whether these returns were produced from a semi-strong efficient or otherwise inefficient market, it is important to exercise caution with the idea that such investment profits might persist. Therefore it is important to be critical towards the suggestion that CSR Alpha may be used to offset losses made via ethical exclusions-based responsible investment strategies.³

In the case of Gompers et al.'s (2003) trading strategy, the Governance Alpha, which disappeared from year 2001 onwards, was initially claimed to be the product of period specific drivers (Core et al., 2005). However this claim was recently and robustly countered by Bebchuk et al. (2010) who proposed that it was the investors' lack of attention to corporate governance information that led to the profitability of the investment strategy in period 1990-2000, and that it was the investors' subsequent learning to incorporate this information in company valuations that caused excess returns to disappear in the following ten year period. They support this claim with their finding that operating performance and Tobin's q continued to be related to governance standards while measures of investor expectations suggested that investors had become less surprised by the association between corporate governance standards and actual operating performance.⁴

In the case of Statman & Glushkov's (2009) CSR Alpha, a similar attention deficit to companies' social and environmental standards (henceforth synonymous with corporate social responsibility (CSR)) is to be expected; it is highly unlikely that period-specific drivers would prevail throughout 1992-2007 and it is equally unlikely that underlying risk factors support abnormal stock returns of top relative to bottom CSR performers as good (poor) CSR is more likely to be risk reducing (increasing) rather than otherwise (Husted, 2005; Logsdon & Wood, 2005; Van De Ven & Jeurissen, 2005).

In this paper, I thus hypothesize that investors have historically given insufficient attention to the full value relevance (Francis & Schipper, 1999; Graham & Dodd, 1934) of CSR information. I suppose that CSR is positively related to operating performance and fundamental shareholder value but that investors have historically overlooked this relationship, and verify this hypothesis by examining stock returns around quarterly earnings announcement date, the accuracy of

² Such as those hypothesized by Core et al. (2006) in their similar examination – and critique – of Gompers, Ishii and Metrick (2003). Core et al. postulate that Gompers, Ishii and Metrick's (2003) finding that companies with weak shareholder rights exhibit significant stock market underperformance was emblematic of "the larger new economy pricing puzzle of the (late) 1990's".

³ Statman & Glushkov (2009) make this claim.

⁴ Investors, despite being able to observe strong empirical evidence that governance information is salient to operating performance, can take time – even a decade according to research supplied in this paper – to properly incorporate publicly available information into their investment processes and resulting expectations.

analysts' forecasts of future operating performance at beginning of fiscal year, and Tobin's q values. I also hypothesize that recent developments in the area of responsible and sustainable investment – such as the launch of the UN PRI and the emergence of various regional and international initiatives by governments and asset owners around the world – helped resolve this attention deficit. I use the year of the launch of the UN PRI as a breakpoint in our analyses to verify this hypothesis. These hypotheses are in line with the idea that stock market efficiency is improving, thus reducing the cost of capital for socially responsible companies – the adverse for socially irresponsible companies – and encouraging more socially responsible corporate behaviour in the economy at large. It implies that CSR's Alpha in the form suggested by Statman & Glushkov cannot be expected to continue to compensate SRI investors for losses made via ethical exclusions.

Findings suggest that investors were significantly differentially more positively surprised at the earnings performance of CSR leaders. Tobin's q moderated the magnitude of this surprise; if investors more fully incorporated the value relevance of CSR information in their assessment of a CSR leaders' prospects (as indicated by higher Tobin's q), then they were less surprised at the actual earnings performance. Finding that CSR is also significantly and positively related to ROA and Tobin's q, these findings suggest that CSR generates superior earnings performance and superior shareholder value, bringing clarity to a forty-year-old debate – on the cause/ effect relationship in CSR's association to measures of operating performance and shareholder value – and disputing the reverse causality explanation based on slack resources theory.

Findings also suggest that investors have gotten better at incorporating the full value relevance of CSR information into their assessment of companies' prospects; CSR Alpha is substantially lower post-2005 while the magnitude of CSR's association to Tobin's q is significantly larger and earnings announcement surprises and analyst forecast errors point to an improvement in stock market efficiency towards CSR information.

Referencing over 500 scholarly articles, Alpha's Tale is firmly grounded in management and capital markets theory and research. It includes a comprehensive review of:

- the strategic management theories underlying the competitive advantage of CSR (section II), and the empirical evidence linking CSR to measures of operating performance (section III) and shareholder value (section IV);
- the empirical evidence that certain trading strategies based on CSR information produces significant risk-adjusted investment profits, capital markets theories underlying possible sources of stock market inefficiency and institutional pressures that support and strengthen this inefficiency (section V);
- the recent developments in the investment industry's institutional environment that lead to an improvement in investors' attention to and treatment of value relevant CSR (ESG)

information into their investment processes, as supported by theories of institutional isomorphism (section VI).

Positioning our contribution among this larger body of work and applying it to actual developments in the investment industry, this paper makes a significant contribution to the fields of strategic management and capital markets practice and research. Hypothesis development, methods and results are addressed respectively in sections VII-IX. I discuss our results in section X and conclude with a summary and final thoughts in section XI. Alpha's Tale applauds the work of certain institutional players such as the UN PRI, which improved stock market efficiency – improving the market's signal to managers and lowering the cost of capital for good CSR companies – and thereby encouraged companies to allocate more resources to CSR.

I discuss the implications of the stock market's markedly inefficient treatment of CSR information in 1992-2005 and the implications of its new appreciation for CSR performance post-2005. Specifically, I discuss how the stock market affects sustainable behaviour by investee companies and either encourages or discourages externalities (either positive or negative). I discuss how this improved market efficiency regarding environmental and social performance will lead to various developments in the investment industry and, more specifically, to new approaches to responsible investment.

I am hopeful that CSR investment profits still exist, but less confident that they exist in the form suggested by Statman & Glushkov. This finding leads us to predict that responsible investing strategies will shift – away from security selection – towards new approaches to strategic asset allocation, 'impact investing' and shareholder engagement. By acting together, asset owners can exercise substantial power over companies with the goal of persuading or requiring the adoption of CSR where such an activity improves shareholder value, and enhances portfolio value. As such, they can contribute to company specific financial performance as well as encourage industry-level best-practices that encourage more sustainable systematic behaviour. Brad Barber (2007) found that CalPERS, one of the largest such investors, created over \$89.5 billion from year 1992 to 2005 in economic benefits by proactively engaging companies to raise their corporate governance standards. Taking account of our findings, the potential for similar engagement to encourage companies to increase resource allocation to CSR seems promising.

It is also foreseeable that such investors might, through shareholder engagement, promote corporate policies that discourage companies from acting in socially irresponsible ways – activities that would have otherwise been net positive on that company's value but net negative on Universal Owners' portfolios. Universal Owners are also expected to increase their engagement with public policy makers to the same end; that is, to use regulatory mechanisms that encourage (discourage) positive (negative) environmental and social externalities from

companies' regular business activities. In this fashion, investors, asset owners and institutional players like the UN PRI are expected to play increasingly important economic and political role.

II. The Competitive Advantage of CSR

The conventional idea that corporate social responsibility can be a competitive advantage (Baron, 2000; Cornell & Shapiro, 1987; Porter, 2006; Solomon & Hansen, 1985) seems to have come a long way since Milton Friedman's (1970) exposé on the topic. And, it seems, so has our definition of the concept:

In his New York Times article, Friedman writes that CSR – which he defines as the use of corporate resources for the good of society, as opposed to shareholder profits – is a corporate expenditure that acts as a tax on investors. He argued that government is better at serving the interests of society and that businesses should therefore only be responsible for the maximization of profit so that – by virtue of this maximand – government contributions would also be maximized and so – indirectly – would the company's contribution to society. According to Friedman, all corporate resources should be allocated in a manner that does not advance any morally legitimate claims of stakeholders other than those of the company's shareholders. This argument rests on agency theory, which explains that agents (corporate executives and managers) might otherwise use their discretionary powers to advance their own individual interest at the expense of shareholders (Fama, 1980; Fama & Jensen, 1983; Jensen & Meckling, 1976) and that all free cash flows should be returned to owners (Jensen, 1986) rather than to wider society (Carroll, 1979).

However, scholars (see Prahalad & Hammel, 1994; Scott, 1995) began to recognize that the success of strategic decisions were highly dependent upon meeting the expectations of various stakeholders (Aguilera et al., 2007; 2003; Aragón-Correa & Sharma, 2003; Bansal & Roth, 2000; Christmann, 2004; Freeman, 1984; Freeman & Evan, 1990; Hill & Jones 1992; Margolis & Walsh, 2003; Sharma & Henriques, 2005; Spar & LaMure, 2003). Managing these stakeholder influences (Astley, 1984; Freeman, 1984; DiMaggio & Powell, 1983; Hakansson & Snehota, 1989; Meyer & Rowan, 1977; Pfeffer & Salancik, 1978) came to be seen as strategic decisions (Eisenhardt & Martin, 2000; Freeman, 1984; Hamel & Prahalad, 1994; Itami, 1987; Prahalad & Hamel, 1990; Teece et al., 1997) that went well beyond the management of traditional industry-based forces identified by Porter (1980). Hence as a form of stakeholder management, strategic 'philanthropy' and corporate donations came to be seen as expenditures that may well be a source of competitive advantage (eg. Fry et al., 1982; Mescon & Tilson, 1987; Wokutch and Spencer, 1987), as opposed to self-interested behaviour by executives (Galaskiewicz, 1997; Haley, 1991): Corporate contribution units could join with other business units to develop donations strategies that increase name recognition among consumers and suppliers, enhance employee morale and productivity, reduce R&D and advertising costs and overcome regulatory obstacles (Smith, 1994; Porter & Kramer, 2002).

Thus, from the idea that philanthropy can be a valuable investment, emerged the thesis that other forms of CSR could also be profitable and coherent with shareholder interests, all while tending to the greater interests of society (Baron, 2000). In short, there was a reorientation of strategy to reflect that “what is good for society does not necessarily have to be bad for the company and what is good for the company does not necessarily have to come at a cost to society” (Moran and Ghoshal, 1996, p. 45). Hence CSR came to be understood according to two subsets of activities and a difference came to be made between CSR that contributes to shareholder value, and CSR that does not (Frederick, 1994): The subset of CSR activities (‘instrumental’ CSR) that attends to primary stakeholders became understood as value building (Berman et al., 1999), while the subset of all other CSR activities (‘altruistic’ CSR) that attends to greater social issues that are not directly related to a company’s primary stakeholders (Hillman & Kleim, 2004) became understood as value destroying and incoherent with the profit maximand (Friedman, 1970). Primary stakeholders, including the natural environment (Starik, 1995), are those that without whose participation the corporation could not survive (Freeman, 1984; Clarkson, 1995). Therefore CSR that tends to those stakeholders could be undertaken with the view of increasing shareholder value (Donaldson and Preston, 1995; Jones, 1995; Jones & Wicks, 1999; Mitchell et al., 1997; Quinn & Jones, 1995) rather than with a primary focus on social good (Carroll, 1999; McGuire et al., 1988).

Managing stakeholder expectations became analogous to managing corporate reputation (Fombrun & Shanley, 1990); by better managing relevant expectations (DiMaggio & Powell, 1983; Freeman, 1984; Pfeffer & Salancik, 1978) in their regular business dealings, companies signal stakeholders about their strategic posture (Fombrun & Ginsberg, 1990; Rumelt, 1974) and transform their track record into a competitively valuable asset (Dowling, 1986; Kreps & Spence, 1985; Murray & Montanari, 1986; Roberts et al., 2002) in the form of superior reputational capital and trust (Barney & Hansen, 1994; Calton & Lad, 1995; Dacin, 1997; Deephouse, 1996; Ring & Van de Ven, 1992, 1994; Suchman, 1995). In that way, instrumental CSR can be thought of as a measure of goodwill from consumers, suppliers, public authorities, and other stakeholders (Rumelt, 1987) like the type of goodwill that occurs from such expenditure as advertising, which presents a company in a favourable light or reduces consumers’ and suppliers’ search for information, for example (Weiss, 1969). As such, a company with a reputation for good CSR can benefit from relatively more stable sales (Milgrom & Roberts, 1986a, 1986b), increased barriers to entry (Comanor & Wilson, 1974), lower transaction costs with suppliers, consumers and public authorities (Craig et al., 2008; Hill, 1995; Jones, 1995; Pfeffer & Salancik, 1978; Westley & Vredenburg, 1991), a unique customer base (Sen & Bhattacharya, 2001), increased customer loyalty (Navaro, 1988), the ability to charge premium prices (Klein & Leffler, 1981; Milgrom & Roberts, 1986a, 1986b), a more efficient workforce (Backhaus et al., 2002; Bhattacharya, 2008; Davis, 1973; Kotter & Heskett, 1992; Pfeffer, 1998; Schuler & MacMillan, 1984; Spence, 1974; Stigler, 1962), etc. As such, a reputation for good CSR can lead to an important competitive advantage (Shrum & Wuthnow,

1988; Weigelt & Camerer, 1988), stratifying companies within industries (Caves & Porter, 1977; Fombrun, 1986), and often translating into a highly valuable ‘license to operate’ or enhancing legitimacy (Baron, 2005; Campbell, 2007; Dacin, 1997; Deephouse, 1996; Dowling & Pfeffer, 1975; Handelman & Arnold, 1999; Kostova & Zaheer, 1999; Post et al., 2002; Selznick, P.: 1957; Sen & Bhattacharya: 2001; Suchman, 1995).

Porter and Kramer (2006) described how companies can use CSR to proactively improve all four main areas of their competitive context (outside-in CSR linkages)⁵ and thereby enhance their prospects – a claim that is supported by much empirical evidence (Bennett, 2002; De Jongh, 2004 ; Gabor, 1991; Hart & Sharma, 2004; Harrison & St. John, 1994; Hillman et al., 2004; Keim, 1978; Legnick-Hall CA. 1996; Marvel, 1977; McWilliams et al., 2002; Oliver & Holzinger, 2008). Regarding companies’ value chain impacts (inside-out CSR linkages), Porter & Kramer (2006) make the assertion that, “No longer can companies be content to monitor only the obvious social impacts of today. Without a careful process for identifying evolving social effects of tomorrow, companies may risk their very survival (p. 84).” This idea provides the platform for ambitious CSR goal setting that builds the valuable firm-specific capabilities to comply with social norms that are above and beyond those required by current industry standards and regulatory frameworks (Hart, 1995; Sharma & Vredenberg, 1998). By doing so, not only can companies enhance their reputation, they can also improve operating productivity (Majumdar & Marcus, 2001) – eg. human capital management that improves worker motivation and performance (Bassi & McMurrer, 2007; Mayo, 2001; Faleye & Trahan, 2006) – and efficiency – eg. lean production methods that reduce costly waste (Porter & Van der Linde, 1995; Romm, 1994), – and can also reap the benefits of early mover advantage regarding future industry standards and regulatory compliance requirements (Arora & Cason, 1996; Dechant et al., 1994; Frynas et al., 2006; Hart, 1995; Shrivastava, 1995b; Marvel, 1977; McWilliams et al. 2002; Porter, 1991; Vredenburg & Westley, 1997).

From these theoretical frameworks and their empirical support, it became obvious that CSR is a meaningful corporate strategy whose payoffs are both deliberate (Robin & Reidenbach, 1988) and emergent (Mintzberg, 1987, 1993), and that are realized by means of two mechanisms: The first is the mere appearance of good CSR, which leads to reputational capital (as seen above); The second is the intrinsic activity of good CSR, which leads to the development of valuable capabilities (Collins & Porras, 1996; Epstein, 1987; Hart, 1995; Kay, 1993; Pruzan & Thyssen: 1990; Sharma & Vredenberg, 1998; Starik & Rands, 1995). These capabilities include a company’s ability to more effectively respond to changes in its competitive context (Barron, 2001; Barnett, 2007; Epstein, 1987; Feddersen & Gilligan, 2001; Hoffman, 1997, 1999; Roome,

⁵ These include the quantity and quality of available business inputs (eg. human resources), the rules and incentives that govern competition (eg. policies that protect intellectual property, ensure transparency and safeguard against corruption), the size and sophistication of local demand (eg. as influenced by product quality and safety standards and consumer rights) and the local availability of supporting industries (eg. service providers and machinery producers)

1994), to enhance its management of risks (Husted, 2005; Kok et al., 2001; Logsdon & Wood, 2005; Schnietz & Epstein, 2005; Van De Ven and Jeurissen, 2005) and reduce the probability of negative events including controversies, consumer or supplier boycotts and lawsuits (Bowie & Dunfee, 2002; Freedman & Stagliano, 1991; Orlitzky & Benjamin, 2001; Peloza, 2006; Story & Price, 2006), while also positioning the company for strategic competitiveness.

Therefore, from a resource-based perspective (Barney, 1991; Penrose, 1959; Wernerfelt, 1984), CSR is ultimately an important source of competitive advantage: Because they are socially complex and causally ambiguous (Branco & Rodrigues, 2006; Hillman & Keim, 2001), the valuable effects of CSR – reputational and otherwise – that improve long-term relationships with suppliers, employees, consumers and other primary stakeholders, and that contribute to cost reduction and revenue opportunities, are difficult for competitors to imitate (Barney, 1986; Barney and Hansen, 1994; Leonard, 1995; Russo & Fouts, 1997; Teece, 1998; Wicks et al., 1999) as they contribute to a unique interplay of human, organizational, and physical resources overtime (Amit & Schoemaker, 1993; Dierickx & Cool, 1989; Lippman & Rumelt, 1982). CSR is thus an intangible asset (Grant, 1991) that provides a sustainable competitive advantage (Barney, 1991; Lado et al., 1992; Lado & Wilson, 1994; Martin et al., 1995; Mudambi & Helper, 1998; Oliver, 1988; Rao, 1994).

In line with this idea, Baron (2000) argued that successful executives position their companies for optimal effectiveness by integrating market strategies with non-market strategies, allowing the company to develop strategic postures from cumulative resource allocation to CSR (Fombrun & Ginsberg, 1990; Rumelt, 1974), thereby doing well by doing good (Paine, 2002; Vogel, 2006). Resource allocation to CSR can therefore be seen as a matter of corporate policy (Copeland & Weston, 1983) that is driven by the identification (Ansoff, 1980; Dutton et al., 1983; Mahon & Waddock, 1992) and evaluation (Bartha, 1983; Bigelow et al, 1993; Dutton & Ashforth, 1993; Dutton & Jackson, 1987) of strategically important CSR issues (Cornell & Shapiro 1987; Edvinsson, 1997; Mayo, 2001; Wood, 1991), and the development and execution of responses to those issues (Greening & Gray, 1993; Post, 1978; Sethi, 1979). As such, a company's level of CSR can be indicative of good management (Andersson & Bateman, 2000; Bansal, 2003; King & Lennox, 2002; Shrivastava, 1995b; Waddock et al., 2002; Waddock & Bodwell, 2004).

To make a parallel (to be useful later in the paper), Mizik and Jacobson (2003) argue that advertising expenses are a suitable proxy for a company's emphasis on value appropriation through the creation of strong brands, while R&D expenses are a good measure of a company's emphasis on value creation through innovation: In this paper, I postulate a similar argument for CSR. The decision to invest in CSR is analogous to the policy to invest in other intangibles such as those accruing from advertising or R&D: Advertising, as discussed previously in this section, provides companies with a measure of goodwill from consumers and other stakeholders, that in turn contributes to differentiating a company's strategic position from that of its competitors

(Fombrun & Shanley, 1990; Rumelt, 1987; Weiss, 1969), and to reducing search costs (Nelson, 1974); R&D contributes to a company's technical capital, which is a factor of production that improves the quality of products or reduces the production costs of existing goods and services (Griliches, 1979), increasing productivity or efficiency (Clark & Griliches: 1984; Lichtenberg & Siegel, 1991), and improving long-run viability (Hitt et al., 1996). CSR is therefore expected to enhance shareholder value just as advertising and R&D expenditures have been shown to do (Ben-Zion, 1978; Canibano et al., 2000; Cockburn and Griliches, 1988; Comanor et al., 1974; Conchar et al. 2005; Griliches, 1981; Hall, 1993; Hall et al., 2005; Hirschey, 1982; Jaffe, 1986; Nelson, 1974; Megna & Klock. 1993; Simon & Sullivan. 1993; Wieser, 2005).

Moreover, CSR, which has been described as a source of 'differentiating innovation' (Kanter, 1999) that is associated with both 'product and process innovation' (McWilliams & Siegel, 2000), builds intangible assets that are positively correlated with (Hull and Rothenberg, 2008; McWilliams & Siegel, 2000; Padgett & Galan 2010; Rothenberg and Zyglidopolous, 2007) and complementary to (Christmann, 2000; Quazi & O'Brien, 2000) those that accrue from expenditures towards other intangibles. As stated by Quazi & O'Brien (2000, p. 35), "the broader dimension of social responsibility, therefore, calls for innovation in production and marketing to reap the benefits of proactive social action". For instance, the reputational benefits from CSR are maximized when related parties are made aware of the activity – advertising is one way to raise awareness – and CSR can be used to reduce search costs in a similar way to advertising; R&D can lead to better CSR – eg. innovating eco-efficient products and processes – and CSR, which encourages greater stakeholder input and consideration into products and processes can lead to R&D (Christmann, 2000).

Most CSR research of the last twenty years has focused on the above outlined business case for CSR. That is the subset of CSR that is in no conflict with Friedman's profit maximand (Lee, 2008). Throughout this paper, I discuss CSR according to the same view. In line with other frameworks that argue that CSR is a key driver of financial performance (Atkinson et al., 1997; Birchard, 1995; Hillman & Keim, 2001; Kaplan & Norton, 1996; Prahalad & Hammel, 1994), I focus on the strategic resource allocation decisions rather than the broader strategy typologies (Miles & Snow, 1978; Porter, 1980) of strategic management scholarship.

With this section's conclusion in mind, I present hypotheses 10-11 (from section VII):

H10: CSR *generates* superior operating performance

H11: CSR *generates* superior shareholder value

III. CSR's Association to Profitability

In their meta-analysis of 167 empirical studies on the CSR/ financial performance relationship, Margolis et al. (2007) find that “after thirty-five years of research, the preponderance of evidence indicates a mildly positive relationship between [CSR] and corporate financial performance (p. 22),” adding that the cross-sectional variation, which is extremely high, is indicative of how complex the relationship is to unravel. Orlitzky et al. (2003), who undertook a similar analysis of the literature, came to the same conclusions, pointing out the necessity for empirical studies to incorporate the important interplay of theory and operational context (Margolis & Walsh, 2001; Rowley & Berman, 2000) in order to unravel the positive association, which is, in many cases, highly significant.

Until the emergence of KLD ratings of CSR in 1990, scholars used a large variety of methods to operationalize the CSR construct, making use of such CSR measures as the Council on Economic Priorities' ratings of environmental performance (Pava & Krausz, 1996; Roberts, 1992), Moskowitz's ratings of social responsiveness (Cochran & Wood, 1984; Sturdivant & Ginter, 1977), CSR perceptions of business leaders, students or business faculty (How Business School Students Rate Corporations', 1972), questionnaire responses by companies (Bagozzi et al., 2001), and various annual report transparency measures (Belkaoui, 1976; Ingram & Frazier, 1983; Freedman & Jaggi, 1982). Later studies made use of a wider variety of increasingly sophisticated indicators, including Fortune magazine's rating of “responsibility to community and environment” (Ballen, 1992; Preston & O'Bannon, 1997; Tichy et al., 1997; Simerly, 1994; Simerly, 1995), charitable donations (Levy and Shatto, 1980; Wokutch & Spencer, 1987; Griffin & Mahon, 1997; Brammer & Millington, 2004; Brown et al., 2006; Fombrun & Shanley 1990; Galaskiewicz, 1997; Roberts, 1992; Seifert et al., 2003), and others. More recent studies tend to make use of CSR ratings by third party agencies such as KLD (Berman et al., 1999; Ruf et al., 2001; Waddock and Graves, 1997).⁶ Our research is based on KLD rating data, which – first published in 1991 – have been described as much more accurate, scalable and credible than those from any other method that came before it (Wood and Jones, 1995).

I begin our comprehensive review of the most important empirical contributions with the first ever published article on the association of CSR to financial performance (Bragdon & Marlin, 1972). Some early, 1972-1992, studies have been criticized for stakeholder mismatching (Wood & Jones, 1995) and for using relatively unsophisticated techniques but are nonetheless informative. These studies, their findings and their subsequent criticisms are a testament to the learning process that occurred throughout the years; a learning process that continues on to this day, as demonstrated by our findings that investors are still unable to fully incorporate the

⁶ Other ratings agencies include Jiantzi Sustainalytics, SiRi, etc. Data providers now include Asset4/Tomson Reuters, Bloomberg and MSCI.

fundamental (Francis & Schipper, 1999; Graham & Dodd, 1934) value relevance of CSR information in their expectations of future operating performance and, thus, stock prices.

-I. Environmental performance

Bragdon and Marlin (1972) were the first to empirically research the CSR/ financial performance link (Margolis et al., 2007). Using the Council on Economic Priorities' pollution performance rankings, which are based on the pollution control records of the largest companies in five highly polluting industries (Council on Economic Priorities, 1977), they constructed a 'pollution control index' composed of the most environmentally friendly companies. Although they failed to control for other variables known to be associated with profitability, they found a positive association between the pollution control index and operating performance defined as EPS growth, ROE and ROC in 1965-1970. A later study found that controlling for company size significantly reduced the relationship to operating performance, although a smaller association did persist (Chen and Metcalf, 1980). It was also shown that the environmental/ operating performance relationship declined in magnitude throughout 1969-1973 (Spicer, 1978a) and that the middle polluters outperformed the best and worst (Bowman and Haire, 1975), suggesting that larger companies are better equipped to profit from pollution control expenditures (Spicer, 1978b) and that there may be a financial penalty for both over- and under-investment in environmental responsibility. Although these early, 'experimental' studies can be challenged on methodological grounds, they are nonetheless symbolic: They suggested a positive link between environmental and financial performance and are an early testament to the importance of controlling for other meaningful predictors of profitability (Wood and Jones, 1995; Russo and Fouts, 1997) – as suggests the somewhat contradictory conclusions of Bowman and Haire (1975).

Other early studies used transparency measures to relate companies' extent of environmental disclosure – defined, for example, as the number of lines in an annual report that are dedicated to the subject of environmental responsibility – to operating performance (Belkaoui, 1976; Ingram & Frazier, 1983; Freedman & Jaggi, 1982). Although their results were inconclusive – finding either an insignificantly small, negative or positive relationship – it marked the beginning of studies based on rating methodologies that aim to assess, from the outside, what is happening inside a company in terms of CSR – usually by the use of company survey/ questionnaires, and focusing on one or a few industries. For example, Bagozzi et al. (2001) obtained company responses to demonstrate that companies with an alignment of managerial commitment, strategic planning and proactive managerial actions toward environmental responsibility are also the ones that achieve the best level of environmental performance – from effective management processes – and, consequently, superior earnings growth and return on investment. Similarly, other studies found that an executive suite's (CEO or equivalent) level of concern towards the environment (Dooley & Lerner, 1994; Judge & Doulgas, 1998; Menguc & Ozanne 2005), which is a proxy for environmental management proactivity, is a significant determinant of profitability. Menguc &

Ozanne (2005) found similar results, but added that environmental initiatives are negatively related to sales growth in the short-term – not so surprising given that environmental responsibility may take more than just a few years to payoff in that regard (ie. reputation may take a few years to sufficiently improve; Hunt and Auster, 1990). Judge and Douglas (1998, p. 256) said it best: “It appears that companies that incorporate concerns for the environment into their strategic planning process possess competitive advantages in the marketplace and are better stewards of the environment.”

Later studies identified meaningful predictors of profitability from environmental performance. King & Lennox (2002), for example, differentiated between environmental responsibility from waste prevention and that which is from waste reduction (treating waste once produced), and found that waste prevention is associated with higher profitability and Tobin's *q* while waste reduction is not. Rothenberg (1999) suggested that these results might be due to synergies that exist between waste prevention and lean production methods (Porter & Van der Linde, 1995; Romm, 1994). Profitability from environmental responsibility is also proven to be linked to the timing of such efforts (Nehrt, 1996), whereby first movers gain a larger profit advantage reflected in stronger net income growth; to industry growth (Russo and Fouts, 1997), which is assumed to be linked to instrumental factors that maximise the reputation benefits from such activity; to company size, wherein smaller companies would disproportionately profit from green economic incentives from governments (Clemens, 2006), and; to pre-existent emission levels, wherein companies with the highest emission levels stand the most to gain (Hart & Ahuja, 1996).

-II. Social performance

Moskowitz (1972, 1975), using his own judgment to identify leading companies, was the first to attempt to measure social performance. Although his method was highly susceptible to his own perceptual biases and thus inherently subjective, the basis for his ratings was correct; he aimed to measure companies' *reputation* for good social performance. Later social performance ratings improved upon Moskowitz' method by more accurately specifying assessment criteria and by increasing the number of individual opinions surveyed ('How Business School Students Rate Corporations', 1972; Ballen, 1992).

Studies based on the initial Moskowitz ratings (Cochran & Wood, 1984; Sturdivant & Ginter, 1977), however, find only a marginally significant relationship to operating performance after controlling for industry and company age, and suggest, similarly to Russo and Fouts (1997), that younger companies are more responsible because they incidentally employ relatively state of the art facilities, operate under relatively modern corporate policies, and also have more flexibility to adapt to social change

Studies based on other methodologies corroborate that controlling for size and industry is important to find a relationship to profitability (Heinze, 1976); the favourable opinion of stakeholders – such as industry regulators, consumers, and other industry members – concerning their satisfaction of a company’s stakeholder communications strategy is positively related to profitability (Starik, 1990), and; that a company’s reputation for customer satisfaction mediates the relationship between CSR and profitability, implying that CSR’s potential to increase customer satisfaction is an underlying mechanism through which the financial promises of CSR are capitalized (Luo, 2006).

Numerous studies have found that Fortune’s rating of a company’s reputation for superior “responsibility to the community and environment” (Ballen, 1992) – as perceived by about 8,000 executives, directors, and market analysts per year – are significantly related to superior profitability (Preston & O’Bannon, 1997; Tichy et al., 1997; Simerly, 1994; Simerly, 1995), while these measures of reputation have also been shown to be highly related to KLD environmental and social ratings (Brown and Perry, 1994): A company’s CSR performance, as measured by KLD, is positively and significantly related to its overall corporate reputation and to its attractiveness as employer; the CSR performance, corporate reputation and employee attractiveness of these companies are all positively and significantly related to profitability and to each other (Turban and Greening, 1997).

Of particular note, it has been shown that the eight qualitative attributes rated in the survey are highly correlated, whereby “the eight attributes elicited from respondents were components of an underlying and stable construct of reputation” (Fombrun & Shanley, 1990, p. 245; Slater and Brown, 1988). Financial performance measures were shown to explain 42 percent (McGuire et al., 1990, p. 176) to 53 percent (Fombrun & Shanley, 1990, p. 250) of the variance of the overall rating. As a remedy, Brown and Perry (1994) used a controlled ‘halo-removal’ technique whereby Fortune attributes were regressed against an index representing recent financial performance. Doing so brought the halo-removed attributes closer to “[KLD’s] objective measures of a corporation’s social responsibility (Brown and Perry 1994, p. 1353)” than was achieved using factor analysis to remove the correlations among the Fortune attributes (Dillon et al., 1984; Fombrun & Shanley, 1990). These results imply the possibility that superior profits leads to above average investment in CSR, rather than the opposite hypothesis that investment in CSR creates a competitive advantage that leads to superior profitability.

-III. Aggregate performance

On aggregate (environmental and social) CSR performance, O’Buehler & Shetty (1976) were among the first to operationalize the idea that “the response of a given company to social demands depends upon social issues which are of particular relevance to its business and upon its own organizational resources and skills (p. 66)”. They surveyed some of the largest companies in the US to uncover internal company variables that are indicative of formal structural attention

to CSR, namely: the establishment of corporate social policies, the creation of a new [CSR] organisational unit, and the election of special interest group representation on the board of directors. They found that such attention to CSR was linked to profitability and influenced by variables such as the size, industry, and ownership of companies; these variables also affected the kinds of problems encountered by the companies in implementing CSR programs. As such, these variables should also be taken into consideration when operationalizing the CSR/ financial performance relationship; They wrote that the “response of a given corporation to perceived social demands depends upon which issues it deems of particular relevance to its business and upon its own organizational resources and skills (p. 78)”;

stated in obvious form, appropriate CSR strategies, given a company’s context, are positively related to operating performance.

Likewise, Goll & Rasheed (2004) write that CSR is a conscious managerial choice “that results from complex internal decision processes (p. 43)” that are tailored to account for the specific economic context of companies and that, “given the significant differences in the properties of the environment from industry to industry and company to company, it seems natural to suggest that the relationship between CSR and company performance may also vary from one environment to another” (p. 44). These ideas are well supported from the research body on the relationship between decision process rationality and company performance (Fredrickson & Mitchell, 1984; Goll & Rasheed, 1997; Priem et al., 1995). Assuming that the CSR/ financial performance relationship is context dependent, they introduce variables that represent the munificence and dynamism of the company’s external environment, as contextual variables that moderate this relationship (Aldrich, 1979). By their and Aldrich’s (1979) understanding, munificence refers to an environment’s ability to support the sustained growth of an organization – in other words it is a measure of the environment’s impact on future cashflows – and dynamism refers to the extent of unpredictable change in an organization’s environment – in other words, a measure of the uncertainty of future cashflows. As would be expected, and similar to Russo and Fouts (1997), they find that CSR is significantly more strongly associated with financial performance in environments that are high in munificence and high in dynamism. CSR would be more profitable for companies whose environment supports high, but relatively uncertain – or volatile – cashflows. This evidence also supports the idea that companies are firmly embedded in their social and economic context. Thus the company that better strategically addresses its operating environment – may it be social, economic, geo-political, etc. – with regards to its own capabilities, would be more likely to outperform financially.

According to Collins and Porras’ (1994) ‘Built-to-Last’ (BtL) argument, the oldest, most profitable, time-tested companies have succeeded as such because – being highly visionary and thus recognizing the need to “weave themselves into the very fabric of society” – they played an instrumental role in the very development of society, while allowing society to play a similar role in the development of the company. Graves and Waddock (2000) applied KLD ratings to these

BtL companies and found that, not only did these top-performers have a long and outstanding record of superior financial performance, but they were also significantly more responsible to five (out of six) KLD-defined stakeholder groups: BtL companies, compared to their non-BtL counterparts, achieved significantly better scores on employee relations, community relations, product (treatment of customer), and diversity measures, while concern for the environment, the last KLD stakeholder group, was marginally significant. Such findings confirm the conclusion, contrary to prior evidence (Aupperle et al., 1985; O'Neill et al., 1989), that a company's executive-suite or Board of Director's level of 'concern for society' – defined by the extent of social forecasting activities, and the presence of a CSR committee – is significantly related to profitability (Waldman et al., 2006).

In the absence of third-party CSR rating agencies other early studies made use of individual indicators as proxies for aggregate CSR performance: Fry and Hocks (1976), which used transparency measures (0.5 points for each photographs related to CSR and 1 point for each paragraph on CSR), found that CSR mentions in annual reports is a poor predictor of profitability, although size and industry classification of companies were significant explanatory variables of CSR disclosure-levels (Cowen et al., 1987; Patten, 1991), while most other studies, which made use of charitable donation-levels, found a positive and significant relationship to following-year operating performance in various industries and geographies (Levy and Shatto, 1980; Wokutch & Spencer, 1987; Griffin & Mahon, 1997; Brammer & Millington, 2004; Brown et al., 2006; Fombrun & Shanley 1990; Galaskiewicz, 1997; Roberts, 1992; Seifert et al., 2003).

Finally overcoming the lack of good and objective CSR aggregate rating assessments, in 1986 the Council on Economic Priorities (CEP; a third-party rating firm) published a one-off comprehensive rating of the CSR activities of 130 major corporations in the US for observation years 1984-1986 (Council on Economic Priorities, 1986).⁷ The rating was based on an extensive analysis by the CEP, and involved: (1) direct communication with each company, (2) a review of corporate annual reports, 10K reports, and proxy statements, (3) an in-depth study of newspapers, magazines, and other publications, and (4) an analysis of secondary information sources such as The Taft Corporate Giving Directory, the National Directory of Corporate Charity, and the National Data Book. Companies were comprehensively rated by a variety of criteria that encompass those used in all previous CSR measures. By evaluating many alternative sources, the CEP provided an improved and more reliable measure of CSR performance. Pava & Krausz (1996) found that companies that were well rated by CEP had superior operating performance relative to a control group of comparable companies in 1985-1991. However, as will be explained in the following sub-sections, it seems that investors did not recognize this relationship until much later (Blackburn et al., 1994). As with other studies

⁷ Interestingly Steven Lydenberg, the main author of the study and Director of Corporate Accountability Research of the CEP, would later co-found Kinder, Lydenberg and Domini & Co. (KLD), where he held the position of Director of Research.

discussed here, it was also found that the more a company was exposed to stakeholder (eg. government and shareholder) influences, the more it engaged in CSR, and the better its CEP rating (Roberts, 1992); a company's level of exposure to business risks would mediate the CSR/ financial performance relationship.

Then, in 1988, the establishment of KLD, a special advisory company on CSR performance, did wonders for the field of CSR/ financial performance research. KLD's ratings of CSR performance overcame many of the shortcomings of previous CSR variables used in research: CSR would be accurately, uniformly and more credibly measured across a wide range of companies and for a consistent range of important social issues (see Appendix I). As Graves and Waddock (1994) wrote:

“(1) [KLD] rates each company on multiple attributes considered important to social performance. (2) Because one group is doing the ratings, using an objective set of screening criteria, the ratings are applied consistently across all companies and are replicable. (3) Because KLD operates a service that supplies information to the investment community at large, it rates all of the Standard & Poor's 500 companies. (4) The group doing the rating consists of knowledgeable individuals not affiliated with any of the rated companies or with researchers performing studies. Thus, KLD's scaling process provides unique access to a wide range of consistently rated companies across a number of important social performance attributes (p. 1039).”

KLD is thereby able to generate ratings of companies' CSR performance that are much more accurate, scalable and credible than those from any other method that came before it (Wood and Jones, 1995).

Controlling for company size, risk and industry, Waddock and Graves (1997) found that KLD's aggregate CSR ratings are positively and significantly related to previous and following year operating profits, lending support for the hypotheses that companies with more slack resource availability have a superior ability to engage in CSR, and that companies that engage in CSR are better managed because they make better use of their assets. Controlling for company size, industry and prior year financial performance, Ruf et al. (2001) found that an improvement in KLD's aggregate CSR ratings are positively and significantly related growth in sales for the current and subsequent year, and to superior return on sales and return on equity in the third subsequent year, suggesting that profitability benefits from increasing CSR expenditures, although not immediate, do materialize in the longer term, supporting both.

Also using KLD ratings, Berman et al. (1999) isolated the effects of CSR on company financial performance by including measures of company strategy (capital expenditures, capital intensity,

cost efficiency and selling intensity)⁸ and controlling for operating environment (munificence, dynamism, and industry concentration).⁹ They found that two (of the five) KLD sub-ratings were directly related to profitability, while all five CSR sub-ratings had a significant indirect effect on company performance, confirming the argument that CSR moderates the relationship between strategy and operating performance (Freeman, 1984; Donaldson & Preston, 1995). Incorporating each KLD sub-ratings independently with strategy constructs, they found that the following interactions were significantly related to operating performance: employee relations and efficiency; the product safety/quality and capital intensity, selling intensity, and cost efficiency; diversity and selling intensity, capital expenditures, and cost efficiency; the natural environment and capital expenditures; the community and capital expenditures. Consistent with prior strategy studies (Capon et al., 1990), they also found that munificence was negatively related to operating performance, while industry concentration was positively related. Having found that nine of the interaction terms used in their regression model were significant, they concluded that “the connections among [CSR], strategy, and financial performance are fairly complex (p. 500).”¹⁰

-.IV. Discussion and Summary

40 years of evidence suggests a positive association between CSR and profitability. Some empirical evidence (Fombrun & Shanley, 1990; McGuire et al., 1990; Waddock and Graves, 1997) suggests a reverse causality relationship; it has yet to be found whether CSR causes companies to be more profitable or whether CSR is rather an expenditure that is carried out by more profitable companies that can afford it. In this paper, I seek to shed light on this debate by contributing findings that more directly speak to the directionality of the cause-effect relationship. While many of the reviewed studies found a compelling number of company- and industry-level factors affecting the relationship, the totality of studies indicates that the profitability/ CSR relationship is quite complex.

With this section’s conclusion in mind, I present hypothesis 1 (from section VII):

H1: CSR is positively associated with higher ROA

⁸ As operationally defined by Hambrick (1983) and based on Porter’s (1980) theoretical foundation.

⁹ As operationally defined by Boyd (1990) and based on the theoretical foundation of Dess & Beard, (1984).

¹⁰ In the early 1990’s, Kinder, Lydenberg and Domini & Co. (KLD; then known as a social choice investment advisory firm) began publishing ratings of companies’ performance on CSR dimensions, spurring a large number of such studies and encouraging the start-up of other similar rating companies around the world. Turban & Greening (1997) and Johnson & Greening (1999), for example, attempt to identify the relationship between a KLD rating and company profitability, but – failing to incorporate appropriate controls for meaningful predictors of profitability – fail to find conclusive evidence of a significant relationship to profitability, leading to the conclusion that environmental responsibility does not have to hurt profits. Other studies based on similar third party ratings come to similar results (Mahoney & Roberts, 2004; Galbreath, 2006; Nakao et al., 2007).

IV. CSR's Incorporation into Asset Pricing

The above empirical evidence on the positive association between CSR and operating performance is congruent with the thesis that CSR can provide a competitive advantage that enhances profitability. It also suggests that CSR should be associated with higher shareholder value, which is, fundamentally, a function of a company's growth and profitability (Francis & Schipper, 1999; Graham & Dodd, 1934; Palepu et al., 1996). But because those studies used accounting measures, which are inherently short-term (Briloff, 1972, 1976; Fisher & McGowan, 1983; Hayes & Aberathy, 1980; McGuire et al., 1986; Ouchi, 1980), they tell us little about the long-term value of CSR (Bentson, 1982; Watts & Zimmerman, 1990) and little about the value that CSR creates for shareholders. Studies indicate that, on average, only 25 percent of a company's market value can be attributed to accounting book value, while the remaining 75 percent consists of intangible assets and expectations of future growth (Eccles, 1991; Lev, 2001).

Market-based measures of shareholder value creation are indicative of the market's perception of the probability distribution of future costs and revenues (Shane & Spicer, 1983) and are therefore better at capturing the fundamental value of CSR; they more accurately incorporate the value of future income streams (Lubatkin & Shrieves, 1986; Rappaport, 1992) and are better at capturing the value of intangible assets (Barney, 1991; Dierickx & Cool, 1989; Grant, 1991; Itami, 1987). Market-based measures incorporate, not only the perceived magnitude of future cash flows, but also the risks associated with achieving such profits (Shane & Spicer, 1983). This point is especially relevant given CSR's potential to enhance a company's risk management (Husted, 2005; Logsdon & Wood, 2005; Van De Ven and Jeurissen, 2005). It is also suggested that market-based measures capture the extent to which ethics-/values-based investors penalize companies for what they see as irresponsible social and environmental behaviour (Heinkel et al., 2001; Landier & Nair, 2009). Thus, from both risk management and ethics-/ values-based perspectives, greater CSR can attract investors, thus lowering a company's cost of capital and increasing its market valuation (Mackey et al., 2007).

The first, most widely used method to study the stock market's incorporation of CSR information into stock pricing is the analysis of stock price reactions to public announcement events with CSR information content. The other method involves analysing the association of various shareholder value measures such as Tobin's q and the price-to-earnings ratio to measures of CSR. Studies using any of these two methods rely on the assumption of semi-strong market efficiency (Fama, 1970), which assumes that stock prices at any time fully reflect all available information. In such a market, stock prices adjust rapidly to the information content of news announcement events in proportion to their revision of the company's prospects, which, in such a

market, follows a rational assessment of all publically available information including historical share prices (Ibid).¹¹

-I. Environmental performance

Given the evidence that relatively high polluters are also likely to be less profitable (see section III.I), it is perhaps not surprising that the stock market reacts negatively to the news announcement of companies' relatively high pollution activities; companies who were announced to be high polluters – upon publication of Toxic Release Inventory data – experienced significant declines in their share value (Hamilton, 1995; Konar & Cohen, 1997), reflecting the market's perception that these pollution-related liabilities would affect profitability, in the immediate-or longer-term.¹² These liabilities could be understood as future expenditures to improve environmental efficiency or costs related to future legal action, worsening reputation, consumer boycott, NGO activism, etc.

As would be expected, Muoghalu et al. (1990) found that share prices significantly declined, by an average of 1.2 percent, upon the filing of a lawsuit related to environmental performance. They also found that market losses due to these lawsuits outweighed the small penalty imposed by courts and regulators, and that share price did not significantly increase at the disposition of the suit. Such evidence, assuming semi-strong markets, indicates shareholders' judgement that the magnitude and/ or volatility (riskiness) of future cashflows is negatively affected beyond the immediate legal liabilities imposed by environmental mismanagement. Others (Karpoff et al., 1998) found a statistically significant average loss of 0.85% to companies involved in negative environmental incidents, but also concluded that stock market penalties stemming from environmental incidents are highly variable and not easy to predict,¹³ while further analyses revealed that the average share price impact of companies named as a potentially liable party in a Superfund cleanup effort, although negative, got smaller through the years (Harper & Adams, 1996), suggesting that investors had gotten better at uncovering such information and incorporating it in share prices, prior to the event. Similar conclusions were reached by Karpoff et al. (2005) and Jones & Rubin (2001).

Studies of the impact of positive CSR news events suggest that the market positively values proactive environmental responsibility by companies; the announcement of environmentally-friendly activities – including marketing or developing products that are friendly to the environment, such as those products with fewer emissions or those having a high recycled

¹¹ We explicitly question the assumption of stock market inefficiency in section V.

¹² The implication of this drop in stock price is that investors updated their expectation of future pollution-related expenditures or liabilities – e.g., probability of accidents, likelihood of exposure under other regulatory programs such as Superfund – which would reduce the future company profitability.

¹³ Karpoff et al. (2005) found an empirically supported possible explanation for such asset pricing that implied market efficiency.

content – is found to have a significant positive effect on stock prices at the day of announcement in the Wall Street Journal in the 1982-1995 (Hall & Rieck, 1998). Investors would also reward companies that receive environmental awards for CSR expenditures to areas such as new or redesigned products and processes that minimize adverse environmental impact and improve environmental safety systems: first-time awards were associated with greater increases in market valuation, and awards to companies in dirtier industries were associated with smaller increases were observed for companies in environmentally dirty industries, possibly indicative of market skepticism (Klassen & McLaughlin, 1996). In 1985-1991, the average award was associated with a \$80.5 million increase in equity value, representing the market's perception of the net present value of future cash flows related to strong environmental performance, whereas investment in environmental management systems has been shown to reduce a company's perceived market risk and thus increase its stock price by about 5 percent (Feldman et al, 1997).

Similar to operating performance studies (Belkaoui, 1976; Ingram & Frazier, 1983; Freedman & Jaggi, 1982) that use transparency measures to assess, from the outside, what is happening inside a company in terms of orientation towards overall environmental performance, several event-studies make use of comparable transparency measures as predictors of the impact of announced environmental regulations on company value (Blacconiere & Northcut, 1997; Blacconiere & Patten, 1994). For example, an examination of stock price reactions to Union Carbide's 1984 Bhopal chemical leak – which caused a significant overall negative market reaction for a sample of 47 companies with chemical operations (other than Union Carbide) as the event was expected to affect future regulatory costs – found that companies with more extensive disclosures of “safeguards to prevent accidents, compliance with current regulations, or general rhetoric concerning environmental awareness” experienced less negative share price declines (Blacconiere & Patten 1994). Similarly, upon the announcement of SARA¹⁴, affected chemical companies with more extensive ex-ante environmental disclosures in their 10-K reports had a significantly less negative reaction to the announcement of the environmental regulation, suggesting that investors interpreted extensive environmental disclosures as a sign that a company is better positioned to manage its exposure to those regulatory costs (Blacconiere & Northcut, 1997). It was also found that those chemical companies with greater exposure to future Superfund costs (based on EPA's estimate of cost liabilities for Superfund sites) had more negative market reaction to SARA (Ibid), supporting the general findings of Barth and McNichols (1994) that EPA data are value relevant; estimates of environmental liabilities

¹⁴ The US' Superfund Amendments and Reauthorization Act was amended in 1986: Among the provisions, SARA increased Superfund funding, increased direct taxes affecting chemical companies, and expanded regulatory disclosure requirements for companies that release hazardous materials into the environment. Thus, SARA increased regulatory costs (particularly for the chemical industry).

provide explanatory power incremental to recognized assets and liabilities in explaining companies' market values).¹⁵

When the Council for Economic Priorities (CEP; a type of third-party rating firm) released its reports on the pollution-control standards of companies, low rated companies experienced significantly more negative returns than companies with high rankings (Shane & Spicer, 1983; Stevens, 1984), indicating the market's learning about possible future regulatory costs and a tarnished reputation for lagging companies. Corroborating this idea that companies are penalized (rewarded) by investors for negative (positive) environmental performance in excess of direct costs: Khanna et al. (1998) found that companies whose environmental performance worsened over time relative to other companies – by measure of their production and treatment of toxic materials – experienced significant declines in share price upon the announcement of results from EPA's TRI assessments through years 1990-1994; Yamashita et al. (1999) found that companies whose Fortune environmental management rating improved does not appear to be a very important issue to stock investors unless improvement of environmental performance scores seems to give the stocks small positive gains, in which case the performance comes as a surprise and investors re-evaluate share value. These findings suggest that investors benchmark companies' historical environmental performance and project it into the future as an input into company valuation; they would thus revise their valuation of share prices at the publication environmental assessments that are contrary to their projections.

Aside from CSR-related news announcement events, there exists evidence that the stock market does incorporate the environmental standards of companies in their overall assessment of companies' value. Konar and Cohen (2001) find that the intangible asset value (Tobin's q) of S&P 500 companies, after controlling for traditional explanatory factors, are negatively related to bad environmental performance (as proxied by toxic chemicals emitted per dollar of revenue, and the number of environmental lawsuits pending). They find that "the average 'intangible liability' for companies in their sample is \$380 million – approximately 9 percent of the replacement value of tangible assets.[...] A 10% reduction in emissions of toxic chemicals resulted in a \$34 million increase in market value (p. 281)", with larger losses accruing to the traditionally polluting industries. Likewise, King & Lennox (2002), who differentiated between emission reduction and prevention, found that more emission prevention is significantly associated with higher future Tobin's q whereas emission reduction (treating waste once produced) is not. It has also been found that S&P 500 multinational companies operating in developing countries achieve higher market values (Tobin's q) by adopting a single, stringent global environmental standard as opposed to defaulting to less stringent, or poorly enforced host country standards (Dowell et al, 2000), suggesting that investors treat environmental externalities as liabilities that the company would eventually have to meet. Companies with established

¹⁵ Holthausen (1994) notes that there are potential specification problems in the Barth and McNichols (1994) balance sheet valuation model.

organizational structures and functions (planning, disclosure, education and management of waste and energy) to implement environmental measures to their negative environmental impact have also been shown have significantly higher Tobin's q values (Nakao et al, 2007). These findings are in line with earlier evidence that companies with good environmental performance also tend to have a higher price/earnings ratio (Spicer, 1978; Chen & Metcalf, 1980), and a higher dividend yield (Greening, 1995).

-II. Social performance

There exists evidence that the stock market punishes companies for socially irresponsible workplace activity in excess of direct costs of litigation (Jarrel and Peltzman, 1985): Significant stock price declines well in excess of legal penalties have been experienced by companies that were announced to have violated employee health and safety standards (Davidson et al., 1994), or equal employment opportunity laws (Hersch, 1998). Such findings reflect the cost of changing workplace standards, or the market's adjustment to information about the company's management and stakeholder reputation effects from bad CSR news announcement events.

Similar evidence exists with regards to corporate fraud and product recalls: Using data on 132 US cases of alleged and actual corporate fraud from 1978 through 1987, it was found that initial news announcement of allegations or investigations of corporate fraud against private parties (suppliers, customers, etc.) or the government were associated with equity value decreases that are "too large to attribute to expectations of impending legal sanctions (p. 759)", wherein just 6.5 percent of lost equity value represents court imposed costs, with penalties and criminal fines accounting for 1.4 percent (Karpoff & Lott, 1993), reflecting the market's perception that future cash flows had been negatively affected by the event. Indeed, it was found that the magnitude of stock price declines, which varied widely across companies, were highly related to the subsequent change in the percentage of a company's revenues, indicating that these companies lost market value in proportion to investors' correct assessment of implied lost future revenues from affected stakeholders that would subsequently lose favour with the company due to being directly affected by the fraudulent activity, and not necessarily in proportion to the direct legal costs of the activity (Karpoff et al., 2005). Similarly, Dowdell et al. (1992), who studied industry effects of the Tylenol poisoning incident, concluded that companies were punished for "indirect costs or costs of goodwill, which would have a greater effect on company value than implied by the immediate outflow of funds resulting from regulation (p. 299)." Such conclusions have also been reached regarding companies' engagement in false advertising (Peltzman, 1981) and other forms of corporate crimes (Reichert et al., 1996), and those that were subject to voluntary and government mandated product recalls (Davidson & Worrell, 1992; Hoffer et al., 1988; Jarrell & Peltzman, 1985).

Research on workplace diversity (Wright et al., 1995) suggests, as expected, that announcements of court settlements on discrimination at the workplace are associated with significant negative

stock price changes, which represent “the capitalization of negative economic implications associated with discriminatory corporate practices (p. 283)”, while the announcements of quality affirmative action programs are associated with significant positive excess returns, which represent “the capitalization of positive information concerning improved business prospects (p. 281)”.¹⁶ These findings support the market’s perception that companies with superior affirmative action practices may be at a competitive advantage – and vice-versa – because of a more talented and committed work force, lower operating costs because of low turnover, low absenteeism and job satisfaction, more creative cultures, good problem solving, high adaptability, and good reputation with diverse customers (Cox & Blake, 1991; Schwartz, 1989).

Likewise, event studies of companies named to the Working Mother Magazine’s list of “Most Family Friendly Companies” (Moskowitz & Townsend, 1994) for the first time – between 1989 and 1994 – found a significant positive abnormal return, in particular for large-cap (NASDAQ) companies¹⁷ (Chauvin & Guthrie, 1994; Jones & Murrell, 2001). These argue that the announcement of the company’s social performance improved the attractiveness of the company as an employer (Turban & Greening, 1997) and that the acknowledgment of the company’s social performance enhanced its overall reputation and image (Dutton & Dukerich, 1991), making it more attractive for investors.

Companies that have better disclosure of health and safety standards also seem to be rewarded by the market: When the US Supreme Court allowed the Occupational Safety and Health Administration (OSHA) to enforce a stricter standard for the reduction of dust emissions in the cotton textile workplace, aiming to protect workers, companies with a higher level of ex-ante disclosures experienced a significantly smaller decline in share price (Freedman & Stagliano, 1991).

Given the above presented evidence, it is not surprising that studies relating stakeholder perceptions of companies’ reputation for good overall social responsibility to measures of company value also find a positive and significant association between the two (Clarkson, 1988; Conine & Madden, 1986).

-.III. Aggregate performance

The set of studies on the market’s incorporation of aggregate CSR information into stock prices is small, but nonetheless revealing.

¹⁶ McWilliams & Siegel (1997) criticized the design of this study; they found – after removing “189 confounding events” – that results were insignificant for a short window immediately surrounding the event.

¹⁷ This confirms their hypothesis that the signalling effect is stronger whenever the signal is widely understood, simple and intuitively appealing, as shown by the difference in outcomes due to the influence of the liquidity and intensity of information processing in which the company’s stock is traded.

An event study revealed that the stock market positively rewards companies upon announcement of charitable donations, as indicated by a significant 1.17% cumulative average abnormal returns from the day of the announcement through 5 days after the event (Hall & Rieck, 1998), implying that the market recognizes the added value from the activity itself, if not the mere publicity of it (Smith, 1994). This stock market reaction is in line with the evidence that companies that donate more to charity have a significantly higher market-to-book ratio (Seifert et al., 2003; Fombrun & Shanley, 1990) and dividend yield (Fombrun & Shanley, 1990).

Companies that engage in a concerted effort to identify and analyze the social and political environment have been shown to have a higher P/E ratio (Newgren et al., 1985) and benefit from the public's perception that the company is a better long-term investment, has a more sound financial position, and makes wiser use of corporate assets (Wartick, 1988). However, Pava & Krausz (1996), who use a CSR dataset provided by the Council on Economic Priorities et al. (1991) find that the market might not fully incorporate the value of CSR for a given company until at least a few years have gone by: Comparing the long-term financial performance of the 53 companies, which were identified as being socially responsible, to a control group of companies matched by both size and industry for 1985-1991, they found that the price to earnings ratio favoured the control group in 1985-1987, but that the effect reversed in the later period to the point that the socially responsible companies had significantly higher market-to-book values in the later, 1989-1991, period (compared to its control group and its 1985-1987 values) – resulting from significantly higher stock returns post-1987 – even though operating performance differential between the two groups of companies remained relatively unchanged throughout 1985-1991. These findings suggest that the stock market incorporated CSR information with a lag.

Interestingly, for S&P 500 companies in 1995-1996, Hillman & Keim (2001) find no significantly positive association between KLD aggregate ratings and Tobin's q and book-to-market ratio – which questions whether CSR is irrelevant for company value, or simply overlooked by the market – although they do find such a relationship to Market Value Added, which is a measure of shareholder value creation that captures the degree of wealth enrichment for the shareholders (Stewart, 1991). I expand on this idea in our study.

-IV. Discussion and Summary

Here again, literature points to a positive association between CSR and measures of shareholder value, but no indication of a conclusive causal link. Do high valuation companies engage in more CSR simply because they have more slack resources to afford it – or – does the very activity of CSR improve a company's prospects and thus its actual market value? Konar and Cohen (2001) echoed these very concerns: “Are highly reputable and profitable companies environmentally sound because they can afford to be, or does that environmental concern enhance their reputation (p. 289)?” At least the stock reactions around news announcement of

companies' positive (negative) CSR activities suggest the market's perception that such activities positively (negatively) contributes to company's prospects.

Given that CSR enhances a company's fundamental value – by its contribution to larger future cashflows, the reduction of risk or some other unknown mechanism – does the market *fully* incorporate CSR value relevant information into stock prices – or – does the market largely overlook such information until much later as suggested by Pava & Krausz (1996)? Hillman & Keim (2001) suggest that investors give insufficient attention to CSR information. Hart & Ahuja (1996), who find that ROE takes longer to be affected by companies' improvement of environmental responsibility than other accounting measures such as ROA and ROS, lend support to the idea that the market is slow at incorporating such information in a company's cost of capital (Barth & McNichols, 1994; White, 1996). Testing for Granger causality (Granger, 1969) between environmental performance and future Tobin's q, King & Lennox (2002) confirmed Hart and Ahuja's (1996) hypothesis: the market took at least a year to incorporate such positive information into company value.

There exists some evidence (Harper & Adams, 1996) that investors learned to better incorporate CSR information into their expectations of company prospects. Even then, do investors *fully* incorporate the fundamental value (Francis & Schipper, 1999; Graham & Dodd, 1934) relevance of CSR information? Posing these questions is analogous to questioning semi-strong market efficiency (Fama, 1965, 1970; Samuelson, 1965); Answering in the negative would suggest that there are abnormal profits to be had for investors who can identify companies affected by the market's improper incorporation of CSR information and capitalize on the market's subsequent adjustment (Lo, 2005). Indeed, it is possible that the fundamental value relevance of CSR is so difficult to understand that investors have not yet been able to accurately fully incorporate such information into their assessment of stock prices.

With this section's conclusion in mind, we present hypothesis 6 (from section VII):

H6: CSR is positively associated with higher Tobin's q.

V. CSR's Potential to Deliver Investment Outperformance

CSR take two dominant forms: (1) an ethics-/ values-based approach, and (2) a best-in-class approach. Ethics-/values-based investment strategies are based on the belief that the universe of investable companies should be constrained to those companies and sectors that are deemed to pass certain ethics-/values-based screens (Mackenzie & Lewis, 1999).¹⁸ Research shows that such investment strategies tend to underperform relative to conventional benchmarks (Geczy et al., 2005; Teper, 1992; Rudd, 1979) due to foregoing investment profits from screened out industries and companies (Hong & Kacperczyk, 2009; Huppé & Hebb, 2011), and implied diversification losses (Kurtz & DiBartolomeo, 1996; DiBartolomeo & Kurtz, 1999).

Best-in-class investment strategies are based on the idea that information pertaining to companies' CSR performance should be incorporated into traditional investment frameworks in a way that enhances investment profits relative to benchmark (CSR Alpha), or in a way that, at least, does not sacrifice such profits (Huppé & Hebb, 2011; Jeucken, 2001; Labatt & White, 2002). These strategies would typically not screen out entire industries, but, within any industry, cherry-pick those companies with the best risk-return/ CSR profile, wherein special attention is often paid to CSR's potential to contribute to a better risk-return profile than that which the market implies (Kelly, 2004). Supporting this 'inefficient markets' hypothesis, an empirical analysis of 61 socially responsible investment (SRI) funds from 1972 to 2000 found that investment strategies employing a relatively large variety of best-in-class considerations "select from a better subset of companies (p. 1113)" and therefore significantly outperformed (Barnett & Salomon, 2006).

In the same spirit, various studies outline CSR-based investment strategies that significantly outperform the market after adjusting for risk and transaction costs: In their 2005 FAJ article entitled "The Eco-Efficiency Premium Puzzle", Derwall et al. outlined a 'long/ short' investment strategy via the most/ least eco-efficient companies – according to Innovest ratings¹⁹ – that yielded a significant 6 percent p.a. return over 1995-2003 after controlling for market sensitivity, investment style and industry bias, or 4.25 percent p.a. assuming transaction costs of 200 basis

¹⁸ These strategies are tailored to socially minded – ethics-/ values-based – investors who do not mind sacrificing part of their potential investment profits if it allows them to feel better, knowing that their investment income is not associated with 'bad' corporate practices. For example, some neo-conservative investors in the US shun companies with affirmative hiring or GLT policies, while others screen out entire sin industries like tobacco, alcohol and gaming based on the notion that investment profits from these companies are linked to negative social consequences (see Statman, 2010).

¹⁹ Innovest uses environmental performance measures as leading indicators for management quality and long-term financial performance. At the heart of Innovest's analytical model is the attempt to balance the level of environmentally driven investment risk with the companies' managerial and financial capacity to manage that risk successfully and profitably into the future.

points. Edmans (2010) found that a value-weighted portfolio of Fortune's 100 Best Companies to Work For in America earned an annual alpha of 3.5 percent from 1984-2009 after controlling for market sensitivity and investment style, and 2.1 percent after controlling for industry. Kempf & Osthoff (2007) found that a 'long/ short' equity strategy via the best-and worst-in-class companies, by KLD's aggregate CSR ratings, produced a positive alpha of 8.7 percent p.a. in 1992-2004 after controlling for market sensitivity and investment style. In their 2009 FAJ article, Statman & Glushkov, who also used KLD ratings and a similar long/short equity strategy – with a slightly different CSR ranking methodology – found a 6.12% p.a. alpha for 1992-2007 after controlling for market sensitivity and investment style.

The presence and magnitude of these significant, persistent risk-adjusted investment profits suggest market inefficiency (Fama, 1970). In a follow up to Derwall et al.'s (2005) FAJ article it was demonstrated that investors' learning to incorporate eco-efficiency information meant that stock prices were adjusted with a lag so that eco-efficient companies started selling at a premium (measured by Tobin's q) by the end of the period under analysis (Guenster et al., 2010), reflecting the market's eventual learning to more fully incorporate the economic value of eco-efficiency. According to these findings, the abnormal returns from Derwall et al.'s (2005) investment strategy are due, in large part, to this price correction by the market. Likewise, Edmans (2010) demonstrated that 1.2-1.7 percent of his strategy's 3.5 percent p.a. risk-adjusted alpha occurred within a 3 day time window centered on earnings announcement date, suggesting that investors misevaluated the fundamental importance of employee satisfaction for future operating cash flows (Bassi & McMurrer, 2007; Mayo, 2001; Faleye & Trahan, 2006).

Similar conclusions have been reached by previous studies that show evidence of an initial market under reaction to information on intangibles (Lev 2004, Daniel and Titman 2006) including R&D and advertising (Chan et al. 2001; Lev and Sougiannis, 1996), and corporate governance standards (Bebchuk et al., 2010) despite evidence that they are value relevant. These results are congruent with La Porta et al.'s (1997) finding that positive earnings announcement surprises account for an important proportion of the outperformance of value over glamour portfolios, especially given post-earnings announcement drift (Ball & Brown, 1968; Bernard & Thomas, 1989), and are supported by evidence that investors rarely attend to information with longer-term implications (DellaVigna & Pollet, 2005; Miles, 1993). Since CSR expenditures can take years to be reflected in earnings (Bentson, 1982; Hunt and Auster, 1990; Menguc & Ozanne, 2005; Ruf et al., 2001; Watts & Zimmerman, 1990), this notion is especially relevant. Given that mispricing exists, investors are expected, however, to eventually learn to incorporate the full value relevance of intangibles information (Lo, 2005).

An alternative explanation to the CSR mispricing hypothesis is the idea that intangibles may be related to an unobservable variable such as a period-specific return driver or an underlying risk factor (Holland, 1986). Core et al. (2006) suggest such an explanation regarding the profitability

of a trading strategy based on governance information (Gompers et al., 2003) although Bebchuk et al., (2010) later found results that suggested mispricing. Edmans (2010), although rejecting the hypothesis, proposed that “employee satisfaction and returns might arise because a third unobservable variable causes both”; take good management, for example (Bloom et al., 2006) – ie. good managers might overinvest in CSR. The evidence supplied by scholars (Campbell et al., 1997; Brown et al., 1996; DeBondt & Thaler, 1995; Ikenberry et al., 1995; Michaelis et al., 1995; Peters, 1993a, 1993b), in line with what I put forward in this paper, points to stock market inefficiency (Jensen, 1978; Shleifer, 1999; Shefrin, 2002; Shiller, 2000).

As stated elsewhere in this paper, the valuable effects of CSR are socially complex and causally ambiguous (Branco & Rodrigues, 2006; Hillman & Keim, 2001) and it is for that reason that they provide a competitive advantage for leading companies (Barney, 1986; Barney and Hansen, 1994; Leonard, 1995; Russo & Fouts, 1997; Teece, 1998; Wicks et al., 1999). The complexity of and the resulting uncertainty towards the relationship to financial performance (Bennett et al., 2005; UN Global Compact, 2004; UNEP, 2001) might also be the reason for investors’ inability (Simon, 1957; Radner & Rothschild, 1975) to correctly incorporate the full value of CSR into their assessment of companies’ earnings prospects and securities pricing (Akerlof, 1970; Grossman & Stiglitz, 1980; Kurz, 1999). This bounded rationality affects multiple levels of the investment market system (Corner et al., 1994; Crossan et al., 1999).

Yet another explanation for CSR mispricing is based on behavioural biases and agency-principal conflicts that result from various institutional pressures (Eisenhardt, 1988, 1989; Perrow, 1968), which are especially relevant given the uncertainty towards the CSR/ financial performance link and the ensuing fiduciary responsibility (Arrow, 1971; Kahneman & Tversky, 1979; Pauly, 1968; Wilson, 1968): The works of Tversky & Kahneman (1972), Makridakis (1990) and Shefrin (2002) provide a good overview of those behavioural biases that could affect investors’ incorporation of CSR into mainstream investment frameworks. These biases – given the historical conventional belief (Boyer & Orléan, 1992) that CSR is solely relevant to ethics-/values-based, as opposed to financially-driven, investors (Cowton, 1994, 1998; Sparkes, 1995, 2001; Hummels & Timmel, 2004) – created a resistance to changing their approach (Abrahamson & Fombrun, 1994; Bettis & Prahalad, 1995; Bjorkman, 1989; Janis, 1972; Johnson, 1987; Kiesler, 1971; Lorsch, 1986; Prahalad & Bettis, 1986; Reger et al., 1994; Stinchcombe, 1965). Again, these apply to multiple levels of the investment market system (Corner et al., 1994; Crossan et al., 1999).

The work of Guyatt (2005, 2008) provides a good overview of those principal-agent challenges – to incorporating value-relevant CSR information – that influence multiple levels of the investment market system; challenges that exist because of institutional pressures (Badrinath et al. 1989; Coates & Davis, 1995; Del Guercio, 1996; Scharfstein & Stein, 1990) that over-emphasize short-term results (Eccles et al., 2001; Graves & Waddock, 1990; Jacobs, 1991;

Porter, 1992), and because of conditions that discourage collaborative action (Ouchi, 1980; Oye, 1986; Williamson, 1985). Short-termism causes the market to overlook the fundamental value of CSR (Juravle & Lewis, 2008) – especially since payoffs to CSR are relatively long-term – thus leading to the mispricing (Black & Fraser, 2000; Cuthbertson et al., 1997; DeLong, et al., 1990; Frederick et al., 2002; Froot et al., 1992b) of CSR. Industry-level collaborative action on CSR incorporation – although it has historically been largely absent, with the exception of the last few years (as seen in the next section) – would have reduced the costs of the activity and improved market efficiency (Guyatt, 2008; Hoffman et al., 1999; Schmid, 2004).

Attention is influenced by peoples' deliberate strategies and intentions (Yantis, 1998). Investors' lack of attention to value relevant company-specific information is shown to be linked to category learning (Barberis and Shleifer, 2003; Hong & Stein, 2007; Peng, 2005; Peng & Xiong, 2006) wherein investors attribute disproportionately more explanatory power to various asset class categories rather than actual company-specific information.

With this section's conclusion in mind, we present hypotheses 2-3 and 8-9 (from section VII):

H2: CSR is positively associated with larger positive earnings announcement surprises.

H3: CSR is positively associated with larger positive analyst forecast errors.

H8: Tobin's q is negatively related to earnings announcement surprises of good CSR companies.

H9: Tobin's q is negatively related to analyst forecast errors of good CSR companies.

VI. CSR's Recent Penetration Into Mainstream Investing

The phrase and acronym ESG (environmental, social and governance) made its first prominent appearances in two December 2004 publications: the UNEP FI Asset Management Working Group report called “The Materiality of Environmental, Social and Governance Issues to Equity Pricing” and a UN Global Compact report called “Who cares wins: Connecting Financial Markets to a Changing World.” Around 2005, Mercer, the world leading investment consulting company, began using the term and acronym ESG (environmental, social and governance) to refer to the wide array of sustainability factors that contribute to investment performance.

The emergence of the term was a testament that these factors, which had historically been overlooked by the stock market, are important considerations for profit seeking investors, and are therefore not only relevant to the domain of ethics-/ values-based investment products (Richardson, 2011). ESG thus represented an investment philosophy that differentiated itself from the broader considerations of SRI: In another UNEP FI (2004b) report centering on the ESG investment philosophy, financial analysts were advised to demonstrate “material links to business value... [and] avoid moral arguments (as quoted in Richardson, 2011).” The ESG dialogue thus centred on the incorporation of such factors into traditional investment processes (Jeucken, 2001; Labatt and White, 2002) to enhance investment decisions, an idea that took an early form in the Enhanced Analytics Initiative (EAI).

Established in October 2004 by a founding group of 7 of the largest institutional investors, the EAI aimed to improve the supply of ESG analysis in the investment value chain, which was thought to be especially important, albeit absent, at the sell-side analysts’ level (Mercer Investment Consulting, 2006). EAI members pledged to allocate at least 5 percent of their brokerage commissions to the brokers providing the best analysis of ESG issues. The initiative resulted in an almost ten-fold increase in quality sell-side research reports on ESG’s materiality for securities pricing, increasing the number of reports produced to a steady average of 145 p.a. from 2005 onwards from 16 and 24 p.a. in 2003 and 2004, respectively (EAI, 2008; See Appendix II). Encouraging sell-side analysts to produce ESG research effectively improved investors’ ability to incorporate ESG analysis into traditional investment processes.

Also in 2005, the global law company, Freshfields Bruckhaus Deringer released a research report called “A legal framework for the integration of environmental, social and governance issues into institutional investment”. The report had been commissioned by the UNEP FI Asset Management Working Group on the heel of the mounting evidence (UNEP FI, 2004a) that the consideration of ESG issues, “in view of their materiality to equity pricing (p. 3)”, might well be a matter of fiduciary duty (Langbein, 2005; Longstreth, 1986) that investment managers cannot legally ignore: After examining a number of common and civil law jurisdictions in North-

America, Europe, Australia and Asia, the Freshfields Report concluded that “[on the basis that] the links between ESG factors and financial performance are increasingly being recognised [...], integrating ESG considerations into an investment analysis so as to more reliably predict financial performance is clearly permissible and is arguably required in all jurisdictions. (p. 13)”, adding that “A majority of the jurisdictions have legislated (or are expected to do so shortly) to require investment decision-makers, particularly in the pensions context, to disclose the extent to which they take ESG considerations into account (p. 11).” Fiduciaries – investment managers and administrators – are required to be competent (Longstreth, 1986) and to act in their beneficiaries’ sole or best interest (Langbein, 2005).

In essence, there arose the articulation of a clear business case that would make it possible to do both well and good by incorporating environmental and social factors into traditional investment frameworks. As a result, June 2005 saw UN Secretary General Kofi Annan convene the largest institutional investors in the world to create a set of Principles for Responsible Investment (PRI; commonly referred to as UN PRI). The UN PRI was then launched in 2006 as a set of six Principles – voluntary and aspirational guidelines on the incorporation of ESG into traditional investment processes and ownership practices – that address the discord between financial markets, the goals of long-term institutional investors and the goals of the UN (Gifford, 2007).

The UN PRI enjoyed almost instant uptake; signatories represented USD 4 trillion in assets under management (AuM) at the launch to USD 10 trillion in 2007, to USD 22 trillion in 2010. By late 2008, the UN PRI was so successful that EAI merged under the same name. Upon signing these Principles, investors commit to 6 principles on the incorporation of ESG into investment processes and ownership practices in recognition that

“As institutional investors, we have a duty to act in the best long-term interests of our beneficiaries. In this fiduciary role, we believe that environmental, social, and corporate governance (ESG) issues can affect the performance of investment portfolios (to varying degrees across companies, sectors, regions, asset classes and through time).”

Since 2005, the successful uptake of ESG led to the creation of various organizations across the world including the European Federation of Financial Analysts (EFFAS) Commission on ESG, which was founded in October 2007 with the objective of facilitating the integration of ESG aspects of corporate performance into investment processes. Major investment consulting companies such as Mercer and Watson Wyatt^{20,21} also joined the movement; Mercer began offering its PRI Implementation Services, for example. The CFA Institute (2008) published an ESG handbook. Leading investment managers began offering ESG products like Goldman Sachs’ Energy Environment and Social (GSEES) Index – an ESG Index introduced in 2005 –

²⁰ Now Towers Watson.

²¹ “Watson Wyatt Hires for ‘Green’ Investing,” Financial News (November 2007).

and today almost every important investment manager has an ESG offering. The movement also saw the emergence of: new investment managers specializing in ESG products like Al Gore's Generational Investment Management (founded in 2004; began investing client money in 2005), for example; new ESG data providers like Bloomberg, Asset4/Thomson Reuters, Innovest/RiskMetrics/ MSCI, etc.; the Stern Report in 2006, and; many other players in the ESG investment space.

Prior to 2005, what is now known as the ESG investment philosophy was referred to under the socially responsible investment (SRI) or social, ethical and environmental (SEE) umbrella terms (Cowton, 1994, 1998; Sparkes, 1995, 2001; Hummels & Timmel, 2004) – terms that were largely associated with ethics/ values-based investing, rather than the mainly profits-oriented perspective of the ESG ideology – further contributing to the difficulty to mainstream the approach (Amaeshi, 2010; Entine, 2003; Kurtz, 2005; World Economic Forum, 2005). ESG has since then become embedded in the lexicon of investment professionals (CFA Institute, 2008; DVFA/ EFFAS, 2010).

The early success of the UN PRI can be explained by institutional theory (DiMaggio & Powell, 1983; Meyer & Rowan, 1977; Weber, 1978; Zucker, 1977), which has been put forward to explain the rapidity of the uptake of ESG incorporation (see Bourghelle et al., 2009). Institutional isomorphism led investors to converge towards the collective belief (Orlean, 2004, 2005) that ESG considerations need be incorporated in investment frameworks: Developments around the conclusions of the Freshfields Report gave rise to coercive pressures (Guyatt, 2008); Uncertainty surrounding costs and benefits of ESG incorporation led to mimetic change to implement ESG considerations like those professed by the UN PRI and its early signatories (Juravle & Lewis, 2008); The development, promulgation and promotion of ideas and information that supports the ESG philosophy by academic and professional organizations gave rise to normative influences (Guyatt, 2005).

Supporting the hypothesis that investor attention towards CSR increased post-2005, Dumas & Louche (2011) “turn to the media coverage of, and discourse on, RI as a proxy for collective representations regarding RI over time. Through a content analysis of press articles spanning from the mid-1980's to 2010, [their] paper traces a longitudinal coverage of Responsible Investment (p. 2).” Following prior work (Tetlock, 2007; Fang & Peress, 2009), they suggest that “the content of financial news could be linked to investor psychology and sociology (Tetlock, 2007, p.2)”. Surveying financial press as a proxy (Gamson & Modigliani, 1989) “for the financial market's perception of [responsible investing], to begin understanding the slow process of ESG mainstreaming (p. 3),” they acknowledge that the financial press as an information provider and forum for opinions seems to be an essential barometer of the formation of a consensus (Bourghelle et al., 2005) and use time brackets (Langley, 1999) to delineate periods emerging from press data.

They identified five distinct periods. The last of which is the period 2005-onwards, which they describe as the ‘ESG years’; during this period:

“media coverage is characterized by a search for neutrality in its wording, away from any ethical shade. It is also characterized by a split focus on corporate governance as an issue linked to RI on the one hand, and on environmental issues on the other hand (with an important coverage dedicated to climate change)... A salient representation in the 2005-2010 data evolves around the link between RI and financial return, which follows a market driven logic. And the content of this topic is formulated as inconclusive, unclear and complex... Finally, two recent notions linked to the lack of good information are the responsibility of investors to address the lack of information, and the link between information availability and skills and resources of investors who must treat this information.”

In contrast, the four other, previous periods are much more focused on the ethics/values-based motive for ESG incorporation. The first two periods 1985-1991 and 1992-1997 are driven by ethics and the dominant representation is that RI may limit financial returns, while the third and fourth periods (1998-2000 and 2001-2004) are seen as a “transition period during which the word ‘responsibility’ becomes the preferred terminology... [2001-2004] is characterized by the abandon of the term ethical, and corresponds to the introduction of the term ‘SRI’”. Also in 2001-2004:

“demand is perceived as existing in the mainstream market for the first time, from pension funds and big mainstream investors (P4Q1.1; P4Q1.3). This demand is sustained by financial, corporate governance, and environmental scandals (P4Q1.2)... The other key issue... is materiality. The articles convey the idea that some things have an impact on financial data (P4Q3.1), while many other factors are immaterial (P4Q3.3)... Short-termism is a notion that appears in the RI press coverage for the first time in this period, and which immediately takes an important place as a salient representation... If RI is long-term, it means it does not fit the short-term focus of financial markets. But it also means RI can offer the long-term vision missing in financial markets. Finally, it means that any effect of RI (performance or materiality for example) would be visible in the long-term. (P4Q4.4).”

The year 2005, which coincides with the formation and the launch of the UN PRI, seems to have been a monumental year wherein various forces (DiMaggio & Powell, 1983; Meyer & Rowan, 1977; Weber, 1978; Zucker, 1977) had a significant impact on investors’ attention to and, thus, treatment of ESG information. It is therefore expected that investor attention towards to full value relevance of CSR information would have significantly improved post-2005.

With this section's conclusion in mind, we present hypotheses 4-5 and 7 (from section VII):

H4: CSR's positive association with larger positive earnings announcement surprises got smaller following industry developments in the area of ESG incorporation

H5: CSR's positive association with larger positive analyst forecast errors got smaller following industry developments in the area of ESG incorporation.

H7: CSR's positive association with higher Tobin's q got larger following industry developments in the area of ESG incorporation.

VII. Hypothesis Development

Our overarching thesis is that the CSR Alpha from best-in-class trading strategies like Statman & Glushkov's is the product of mispricing (see section V), rather than underlying risk factors or period specific drivers. I thus examine the evidence for such a causality relationship in the CSR Alpha story.²²

As supported by theory that links CSR to competitive advantage (see section II), and by the aggregate empirical evidence put forward (see section III), I hypothesize that CSR is positively associated to operating performance (H1).

H1: CSR is positively associated with higher ROA

In line with our thesis of CSR mispricing (see section V), I hypothesize that investors have historically overlooked the importance of CSR for future operating performance: investors are positively surprised at the earnings performance of leading versus lagging CSR companies (H2; H3), and – as a consequence of recent industry developments that encouraged ESG incorporation (see section VI) – investors learned to more fully incorporate the value-relevance of CSR into their expectations of earnings performance and (H4; H5).

H2: CSR is positively associated with larger positive earnings announcement surprises.

H3: CSR is positively associated with larger positive analyst forecast errors.

H4: CSR's positive association with larger positive earnings announcement got smaller following industry developments in the area of ESG incorporation

H5: CSR's positive association with larger positive analyst forecast errors got smaller following industry developments in the area of ESG incorporation.

Also supported by theory that links CSR to competitive advantage (see section II), and by the aggregate empirical evidence put forward in this paper (see section IV), I hypothesize that CSR is positively associated to market-based measures of shareholder value (H6), and that – since investors had historically overlooked the full fundamental value-relevance of CSR – these market-based measures are more positively associated to CSR following industry developments that encouraged ESG incorporation (H7; see section VI).

²² Given that these returns are indeed due to inefficiency, investors should exercise caution with the idea that such returns would persist in the future because, as required by the adaptive markets hypothesis (Lo, 2005), market inefficiencies are expected to disappear as investors learn to arbitrage the information.

H6: CSR is positively associated with higher Tobin's q.

H7: CSR's positive association with higher Tobin's q got larger following industry developments in the area of ESG incorporation.

Implied by premises leading to H3, H5 and H7, I hypothesise that market-based measures of (fundamental) shareholder value moderates CSR's association to earnings announcement surprise and analyst forecast errors; the market-based measure of shareholder value (Tobin's q) is an indicator of the extent to which investors actually incorporate the value relevance of CSR information into their assessment of a company's prospects. If investors more fully incorporated CSR information into their assessment of a company's prospects, the less surprised they should be at the company's actual operating performance (H8; H9).

H8: Tobin's q is negatively related to earnings announcement surprises of good CSR companies.

H9: Tobin's q is negatively related to analyst forecast errors of good CSR companies.

On the positive association between CSR and measures of financial performance – if our findings support H2, H3, H8 and H9, then they support the hypothesis that CSR caused superior operating performance and superior shareholder value (Joshua et al., 1996): That investors are positively more surprised at the earnings performance of good CSR companies suggests that CSR *causes* superior operating performance; That investors are less surprised at the earnings performance of good CSR companies if they already have more fully incorporated the value relevance of CSR information into their assessment of shareholder value suggests that CSR *causes* superior (fundamental) shareholder value.

H10: CSR *generates* superior operating performance.

H11: CSR *generates* superior shareholder value.

VIII. Methods and Methodology

I limit our analyses to S&P 500 constituents as do several other studies (Dowell et al, 2000; Hillman & Keim, 2001; Konar & Cohen, 2001). This allows us to focus on those companies that are most intensely followed by investors and that are also more comprehensively covered by KLD throughout 1991-2009.

-I. The CSR variable

I use the same CSR data and ranking methodology as Statman & Glushkov (2009): The CSR data is from KLD Research and Analytics.²³ Each company in KLD's database is rated by its social and environmental profile according to six characteristics: community, employee relations, diversity, environment, products, and human rights.²⁴ KLD's determination of strengths and weaknesses follows a extensive proprietary process that includes a comprehensive examination of publicly available information²⁵ to cover approximately 80 indicators.²⁶

A company's CSR score by characteristic is determined by subtracting the weaknesses from the strengths. This score is then industry-adjusted by subtracting the characteristic mean score of all KLD rated companies in its industry that year, using the 20-industry SIC-based classification of Moskowitz and Grinblatt (1999). As opposed to Kempf & Osthoff (2007), companies for which KLD attributed no strengths nor weaknesses for the given characteristic are excluded from these 'best-in-class' calculations. Companies are then ranked and the top third companies are put in a 'best-in-class' bucket while those in the bottom third are put in the 'worst-in-class' bucket. This procedure is repeated for all six characteristics.

If a company is in the 'best-in-class' bucket by two or more CSR characteristics and not in the 'worst-in-class' bucket by any CSR characteristic, then it is a good CSR company (CSR+). If a company is in the 'worst-in-class' bucket by two or more CSR characteristics and not in the 'best-in-class' bucket by any CSR characteristic, then it is a bad CSR company (CSR-). Statman & Glushov (2009) used this methodology to form year-end portfolios, calculating the returns of an equally weighted portfolio that was long the stocks of the good CSR companies (CSR+) group by the environment characteristic and short the stocks of the bad CSR companies (CSR-).

-II. CSR's association to operating performance

²³ Now part of MSCI.

²⁴ KLD also rates companies on their governance standards, but, like Statman & Glushkov (and Kempf & Osthoff (2007)), we do not include this characteristic into our ranking methodology.

²⁵ At no point does KLD engage with actual companies for data collection .

²⁶ These indicators cover the panoply of CSR issues incorporated in all previous CSR measures used in academic studies reviewed in sections III and IV. For a list of KLD strength and weakness indicators by characteristic, see Appendix I.

Return on assets (ROA) is calculated as the ratio of earnings before depreciation (EBITDA) to the average of beginning- and end-of-year total assets. Like Bebcuk et al. (2010), the ROA_t variable is the natural logarithm of industry adjusted ROA; ROA is industry adjusted by dividing it by the industry median of S&P 500 constituents by the Fama & French (1997) 48 industry classification.

$$(1) \text{ ROA}_t = \alpha + \beta_1 \text{CSR}_{t-1} + \beta_3 \log(\text{Assets})_t + \beta_4 \log(\text{Age})_t + \beta_5 \log(\text{BME})_{t-1} + \beta_6 \text{Capex}_t + \beta_7 \text{ADV}_t + \beta_8 \text{RND}_t + \beta_9 \text{IndCapex}_t + \beta_{10} \text{IndADV}_t + \beta_{11} \text{IndRND}_t + \beta_{12} \text{GOV}_{t-1} + \beta_{13} \text{ADV_Dummy}_t + \beta_{14} \text{RND_Dummy}_t + \beta_{15} \text{IndADV_Dummy}_t + \beta_{16} \text{IndRND_Dummy}_t + \beta_{17} \text{Year_Dummy}_t + \beta_{18} \text{FF48_Dummy}_t$$

CSR_{t-1} is a dichotomous variable that takes the value of 1 if the company is a good CSR company (CSR+) and 0 if it is a bad CSR company (CSR-). If β_1 is significantly positive (negative) then our interpretation is that CSR performance is associated with higher (lower) operating performance.

As controls, I include the log of total assets, the log of age, the log of the book-to-market ratio and the company's governance rating to account for that variable's importance for operating performance, as demonstrated by prior literature (Bebchuk et al., 2010; Core et al., 2006), and to account for the possibility that a company's level of governance (managerial entrenchment) might influence its decision to invest, or not, in CSR (Andersson & Bateman, 2000; Bansal, 2003; King & Lennox, 2002; Shrivastava, 1995a; Waddock et al., 2002; Waddock & Bodwell, 2004). I also control for indicators of company-level characteristics of strategy orientation with with industry-adjusted capital capital expenditure (Capex_t) as an indicator of asset parsimony and with industry-adjusted capital advertising intensity (ADV_t) and industry-adjusted capital R&D intensity (RND_t) as indicators of differentiation.

As suggested by McWilliams & Siegel (2000) and Fisman et al. (2007), I include the following industry-level controls: industry Capex (IndCapex_t), industry advertising intensity (IndADV_t) and industry R&D intensity (IndRND_t). These industry-level controls are defined as the mean value of industry S&P 500 constituents.

As with equations 1-4: the governance variable (GOV_{t-1}) is Bebcuk et al's (2009) governance proxy, which they call the entrenchment index; all accounting data was obtained from Compustat; all industry adjusted controls are adjusted by dividing by the S&P 500 industry median according to the Fama & French (1997) 48 industry definitions; industry level controls are calculated as the industry average of S&P 500 constituents according to the Fama & French (1997) 48 industry definitions; I include dummies for missing ADV_t , RND_t , IndADV_t and IndRND_t , year and industry, and; age is calculated in months by subtracting the "beginning of

stock data” year from the CRSP header information dataset from the actual year of observation and multiplying by twelve.

As with equations 1-4: advertising intensity is calculated as the ratio of advertising expenditures to sales; R&D intensity is calculated as the ratio of advertising expenditures to sales, and; Capex is calculated as the ratio of capital expenditures to total assets.

-III. CSR’s incorporation into asset pricing

Cumulative abnormal returns (CAR_t) are calculated for windows -20:+1, -10:+1, -5:+1, -3:+1 and -1:+1 around actual quarterly earnings announcement dates and are benchmarked against the value-weighted Fama & French (1991) three-factor model augmented by Carhart’s (1997) momentum factor. Quarterly earnings announcement dates are obtained from I/B/E/S. Following prior work (Bebchuk et al, 2010; Giroud and Mueller, 2011), I limit this analysis to companies that are followed by at least 5 analysts upon the last I/B/E/S statistical period indicator before actual earnings announcement.

$$(2) \text{ CAR}_t = \alpha + \beta_1 \text{CSR}_{t-1} + \beta_2 Q_{t-1} * \text{CSR}_{t-1} + \beta_3 Q_{t-1} + \beta_4 \text{ADV}_{t-1} + \beta_5 \text{RND}_{t-1} + \beta_6 \text{GOV}_{t-1} + \beta_7 \text{Post01}_t + \beta_8 \text{Post01}_t * \text{GOV}_{t-1} + \beta_9 \text{Post05}_t + \beta_{10} \text{Post05}_t * \text{CSR}_{t-1} + \beta_{11} \text{IndADV}_{t-1} + \beta_{12} \text{IndRND}_{t-1} + \beta_{13} Q_{t-1} * \text{ADV}_{t-1} + \beta_{14} Q_{t-1} * \text{RND}_{t-1} + \beta_{15} Q_{t-1} * \text{GOV}_{t-1} + \beta_{16} \text{ADV_Dummy} + \beta_{17} \text{RND_Dummy} + \beta_{18} \text{IndADV_Dummy} + \beta_{19} \text{IndRND_Dummy} + \beta_{20} \text{Year_Dummy}_t + \beta_{21} \text{FF48_Dummy}$$

Forecast errors are calculated by subtracting the analyst consensus median forecast at 8 months before fiscal year end from the actual earnings performance. These data are obtained from IBES. Errors are scaled by the previous year’s total assets, and previous year’s price per share as reported by Compustat. Following prior work (Bebchuk et al, 2010; Giroud and Mueller, 2011), I limit this analysis to companies that are followed by at least 5 analysts eight months before fiscal year end.

$$(3) \text{ Forecast_Error}_t = \alpha + \beta_1 \text{CSR}_{t-1} + \beta_2 Q_{t-1} * \text{CSR}_{t-1} + \beta_3 Q_{t-1} + \beta_4 \text{ADV}_{t-1} + \beta_5 \text{RND}_{t-1} + \beta_6 \text{GOV}_{t-1} + \beta_7 \text{Post01}_{t-1} + \beta_8 \text{Post01}_{t-1} * \text{GOV}_{t-1} + \beta_9 \text{Post05}_t + \beta_{10} \text{Post05}_t * \text{CSR}_{t-1} + \beta_{11} \text{IndADV}_{t-1} + \beta_{12} \text{IndRND}_{t-1} + \beta_{13} Q_{t-1} * \text{ADV}_{t-1} + \beta_{14} Q_{t-1} * \text{RND}_{t-1} + \beta_{15} Q_{t-1} * \text{GOV}_{t-1} + \beta_{16} \text{ADV_Dummy} + \beta_{17} \text{RND_Dummy} + \beta_{18} \text{IndADV_Dummy} + \beta_{19} \text{IndRND_Dummy} + \beta_{20} \text{Year_Dummy}_{t-1} + \beta_{21} \text{FF48_Dummy}$$

Again, CSR_{t-1} is a dichotomous variable that takes the value of 1 if the company is CSR+ and 0 if it is CSR-. If β_1 is significant and positive (negative) then our interpretation is that CSR performance is associated with higher (lower) relative earnings announcement surprise.

The Tobin’s q (Q_{t-1}) variable is calculated as in equation 4; it is the natural logarithm of the previous fiscal year’s ratio of Tobin’s q to the industry median Tobin’s q. On its interaction with CSR ($Q_{t-1} * \text{CSR}_{t-1}$), if β_2 is significant and negative (positive) then our interpretation is that

Tobin's q negatively (positively) moderates the magnitude of investor's surprise to companies' actual operating performance.

The post-2005 ($Post05_t$) variable accounts for investors' increased attention to CSR information post-2005 (see section VI). It is a dichotomous variable that takes the value of 1 if the observation year is post-2005 and 0 if it is not. If β_{10} is significant and positive (negative), then our interpretation is that investors learned to more fully incorporate the value-relevance of CSR information into their assessment of companies' future earnings prospects post-2005.

I control for the company's level of governance (GOV_{t-1}) following prior work that shows this variable's relationship to earnings announcement surprises (Bebchuk et al., 2010). I add a post-2001 ($Post01_t$) and the interaction term of that variable with the company's level of governance ($Post01_t * GOV_{t-1}$) to account for investors' learning to more fully incorporate such information into their assessment of companies' future earnings prospects post-2001 as suggested by Bebhuk et al. (2010).

I also add company advertising intensity (ADV_{t-1}), company R&D intensity (RND_{t-1}), industry advertising intensity ($IndADV_{t-1}$) and industry advertising intensity ($IndRND_{t-1}$) as these have been shown to be linked to both a company's CSR performance (Christmann, 2000; Hull and Rothenberg, 2008; Kanter, 1999; McWilliams & Siegel, 2000; Padgett & Galan 2010; Quazi & O'Brien, 2000; Rothenberg and Zyglidopoulos, 2007) and investors' earnings announcement surprises (Chan et al. 2001; Daniel and Titman 2006; Lev, 2004; Lev and Sougiannis, 1996).

To isolate the Q_{t-1} and $Q_{t-1} * CSR_{t-1}$ parameters from the influence of advertising intensity, R&D intensity and governance on investor expectations, I included the interaction terms $Q_{t-1} * ADV_{t-1}$, $Q_{t-1} * RND_{t-1}$ and $Q_{t-1} * GOV_{t-1}$.

-.IV. CSR's association to shareholder value

Following Bebhuk et al. (2010) Tobin's q (Q_t) is calculated as defined by Kaplan and Zingales (1997),²⁷ and is adjusted by its industry median and natural logarithm.

$$(4) \quad Q_t = \alpha + \beta_1 CSR_{t-1} + \beta_3 \log(\text{Assets})_t + \beta_4 \log(\text{Age})_t + \beta_5 \text{Leverage}_t + \beta_6 \text{ROA}_t + \beta_7 \text{Capex}_t + \beta_8 \text{ADV}_t + \beta_9 \text{RND}_t + \beta_{10} \text{IndCapex}_t + \beta_{11} \text{IndADV}_t + \beta_{12} \text{IndRND}_t + \beta_{13} \text{GOV}_{t-1} + \beta_{14} \text{Post01}_t + \beta_{15} \text{Post01}_t * \text{Gov}_{t-1} + \beta_{16} \text{Post05} + \beta_{17} \text{Post05} * \text{CSR} + \beta_{\dots} \text{ADV_Dummy}_t + \beta_{\dots} \text{RND_Dummy}_t + \beta_{\dots} \text{IndADV_Dummy}_t + \beta_{\dots} \text{IndRND_Dummy}_t + \beta_{\dots} \text{Year_Dummy}_t + \beta_{\dots} \text{FF48_Dummy}$$

²⁷ Under this definition, Tobin's Q is the market value of assets divided by the book value of assets plus the market value of common stock less the sum of book value of common stock and balance sheet deferred taxes.

CSR_{t-1} is a dichotomous variable that takes the value of 1 if the company is CSR+ and 0 if it is CSR-. If β_1 is significantly positive (negative) then our interpretation is that CSR performance is associated with higher (lower) Tobin's q values.

As controls, I include the log of total assets, the log of age, industry-adjusted ROA, industry-adjusted leverage, and the company's governance rating to account for that variable's importance for shareholder value, as demonstrated by prior literature (Bebchuk et al., 2010; Core et al., 2006), and to account for the possibility that a company's level of governance (managerial entrenchment) might influence its decision to invest, or not, in CSR (Andersson & Bateman, 2000; Bansal, 2003; King & Lennox, 2002; Shrivastava, 1995a; Waddock et al., 2002; Waddock & Bodwell, 2004). I also control for indicators of company-level characteristics of strategy orientation with industry-adjusted capital expenditure ($Capex_t$) as an indicator of asset parsimony and industry-adjusted advertising intensity (ADV_t) and industry-adjusted R&D intensity (RND_t) as indicators of differentiation. Leverage is calculated as the ratio of long-term and short-term debt to total assets.

As with equation 1, I include industry Capex ($IndCapex_t$), industry advertising intensity ($IndADV_t$) and industry R&D intensity ($IndRND_t$), defined as the mean value of industry S&P 500 constituents. I added industry leverage ($IndLeverage_t$) as it was included as a measure of risk in our Tobin's q regression.

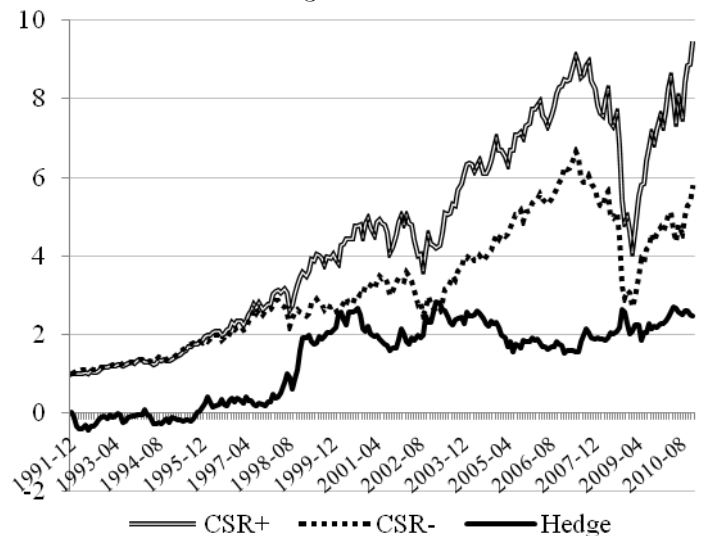
Postulating a learning effect, as with equations 2 and 3, I add $Post01_t$, $Post01_t * GOV_{t-1}$, $Post05_t$ and $Post05_t * CSR_{t-1}$ for the same reasons explained above in the case of earnings announcement surprises and analyst forecast errors. Only, with this regression, if, for example, β_{17} is positive (negative), then our interpretation is that investors positively reassessed the value relevance of CSR information into their assessment of (fundamental) shareholder value post-2005.

XI. Results

Our findings support our overarching thesis that CSR caused stock returns. They suggest that CSR Alpha was product of CSR mispricing, rather than a product of an underlying risk factor or period specific drivers. They also point to investors learning to more fully incorporate the value relevance of CSR information into their assessment of companies' prospects post-2005.

An equally weighted portfolio of the CSR leaders (CSR+) produced a significant 4.32 percent annualized Alpha in period 1992-2005, and an insignificant negative alpha post-2005, while the counterpart portfolio of the CSR laggards (CSR-) produced negligible alphas during the same periods. (See Tables 6-7 in Appendix IV.) Because our analyses are limited to the S&P 500, I do not find – as do Statman & Glushkov – the presence of significant investment outperformance from the hedge strategy that is 50 percent long the CSR+ portfolio and 50 percent short the CSR- portfolio during these periods. These results point to investor's failure to incorporate the full value relevance of information on companies' CSR leadership in period 1992-2005. Focusing on CSR+, I find evidence that risk-adjusted returns decreased substantially post-2005, suggesting that investors got better at incorporating such information into stock prices. (See also Appendix V.)

Figure 1. Cumulative Returns to Trading Strategies (1992-2010)



Notes: Represents the cumulative monthly returns of three portfolios: CSR + (equally weighted long-only portfolio of CSR leaders), CSR – (equally weighted long-only portfolio of CSR laggards) and Hedge (50 percent CSR+ and 50 percent CSR-) for the time period starting Jan. 1992 and ending Dec. 2010.

Table 1. CSR+ Portfolio Multifactor Regression Results

Period	α	$R_m - R_{ft}$	SMB	HML	MOM	Post05	R^2
1992- 2005	4.32% ** (0.001453)	1.0158 *** (0.0359)	-0.0521 (0.0426)	0.2238 *** (0.0608)	-0.1132 *** (0.0342)		0.8478
2006- 2010	-0.09% (0.001283)	1.0391 *** (0.0268)	0.3327 *** (0.0658)	-0.0384 (0.0498)	-0.1421 *** (0.0290)		0.9774
1992-2010	4.29% ** (0.00142)	1.0221 *** (0.0264)	-0.0164 (0.0386)	0.2137 *** (0.0473)	-0.1181 *** (0.0292)	-2.61% (0.0023)	0.8945
1992-2009‡	5.02% *** (0.0015)	1.0064 *** (0.0276)	-0.0288 (0.0391)	0.2111 *** (0.0501)	-0.1244 *** (0.0308)	-4.81% * (0.0025)	0.8926

Notes: Reports a sub-period breakdown of monthly regression results of an equal-weighted long-only portfolio of

CSR+ companies by sub-period. For each year, average alpha is estimated using 12 months of data corresponding to the calendar year. α and Post05 are annualized percentages. White (1980) robust standard errors are used throughout and are reported in parentheses. The Fama-French (1993) three-factor model enhanced with the Carhart (1997) momentum factor was used as a benchmark.

***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

‡Excluding year 2005.

If these performance differentials – CSR+ versus CSR-; 1992-2005 versus post-2005 – were indeed the product of market inefficiency, and not that of an underlying risk factor or other period specific drivers, I would expect to find evidence that analysts did (do) not fully incorporate CSR information into their forecasts of companies' prospects and that investors were (are) positively surprised at the actual earnings performance of CSR+ companies. And I do find such evidence.

Analysts' forecast errors of companies' annual earnings performance were significantly more positive for CSR+ companies (supporting H3), suggesting that investors underestimated CSR's positive contribution to future operating performance. There is some evidence (also see Tables 10-12 in Appendixes VI-VII) that forecast errors got smaller post-2005 (H5).

Table 2: Forecast Errors 1992-2009

Scale	CSR _{t-1}	$\frac{Q_{t-1}^*}{CSR_{t-1}}$	Q _{t-1}	$\frac{Post05_t^*}{CSR_{t-1}}$	Post05 _t	Cons.	Obs.	R ²	Adj. R ²	F Stat	P-Value
Assets _{t-1}	0.0083 *** (0.0024)	-0.0288 ** (0.0118)	0.0144 (0.0139)	0.0041 (0.0049)	-0.0098 (0.0090)	0.0220 (0.0349)	1461	0.1154	0.0648	2.2811	0.0000
PPS _{t-1}	0.0089 ** (0.0042)	-0.0397 * (0.0209)	0.0827 *** (0.0246)	0.0019 (0.0087)	-0.0101 (0.0159)	-0.0109 (0.0615)	1461	0.1275	0.0776	2.5548	0.0000

Notes: Reports the relationship between analyst forecast errors (scaled by Assets_{t-1} and PPS_{t-1}) and CSR performance during 1992-2009 using OLS regression of the announcement returns in a particular window on the terms listed in the table, and GOV_{t-1}, ADV_{t-1}, RND_{t-1}, IndADV_{t-1}, IndRND_{t-1}, Q_{t-1}*GOV_{t-1}, Q_{t-1}*ADV_{t-1}, Q_{t-1}*RND_{t-1}, Post01_t and Post01_t*GOV_{t-1}, which were suppressed for ease of presentation. The regression also included dummies for industry, fiscal year, missing ADV_{t-1}, missing RND_{t-1}, missing IndADV_{t-1} and missing IndRND_{t-1}. An analyst forecast error is Post05_t if the actual analyst consensus forecast was issued in years 2005-onwards, similarly for Post01_t.

***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

As the CSR mispricing story goes, investors' inability to fully appreciate – eg. at the beginning actual fiscal year, as evidenced by analysts' forecast errors – CSR's potential to positively impact a company's earnings potential lead them to upwardly revise their assessment of CSR+ companies' prospects, and thus market value, during following time periods. Results from our analysis of stock price behaviour around quarterly earnings announcement dates support this hypothesis; I find that investors are positively and significantly more surprised at the quarterly earnings performance of CSR+ companies compared to their CSR- counterparts (supporting H2) in period 1992-2010, being less surprised post-2005 (supporting H4; also see Table 9 and 13 in Appendixes VI and IX).

Table 3: Earnings Surprise 1992-2010

Window	CSR_{t-1}	Q_{t-1}^* CSR_{t-1}	Q_{t-1}	$Post05_t^*$ CSR_{t-1}	$Post05_t$	Cons.	Obs.	R^2	Adj. R^2	F Stat	P-Value
-1:+1	0.0041 ** (0.0021)	-0.0122 (0.0010)	0.0080 (0.0117)	-0.0028 (0.0039)	-0.0175 * (0.0093)	0.0015 (0.0111)	6104	0.0219	- 0.0002	0.9916	0.5117
-3:+1	0.003719 * (0.002245)	-0.0150 (0.0108)	0.0105 (0.0126)	-0.0032 (0.0042)	-0.00041 (0.0100)	-0.0009 (0.0119)	6104	0.0240	0.0019	1.0863	0.2357
-5:+1	0.0042 * (0.0024)	-0.0141 (0.0116)	0.0109 (0.0136)	-0.0031 (0.0045)	0.0004 (0.0108)	-0.0048 (0.0129)	6104	0.0318	0.0100	1.4528	0.0006
-10:+1	0.0052 * (0.0029)	0.0034 (0.0141)	-0.0077 (0.0164)	-0.0048 (0.0055)	0.0048 (0.0131)	-0.0071 (0.0156)	6104	0.0480	0.0265	2.2284	0
-20:+1	0.0064 * (0.0036)	0.0377 ** (0.0173)	-0.0347 * (0.0203)	-0.0054 (0.0068)	-0.0037 (0.0161)	-0.0233 (0.0192)	6104	0.0733	0.0524	3.4984	0

Notes: Reports the relationship between earnings announcement returns (windows ranging -1, -3, -5, -10 and -20 to +1 trading days around quarterly earnings announcement dates), and CSR performance during 1992-2010 using OLS regression of the announcement returns in a particular window on the terms listed in the table and GOV_{t-1} , ADV_{t-1} , RND_{t-1} , $IndADV_{t-1}$, $IndRND_{t-1}$, $Q_{t-1}^* GOV_{t-1}$, $Q_{t-1}^* ADV_{t-1}$, $Q_{t-1}^* RND_{t-1}$, $Post01_t$ and $Post01_t^* GOV_{t-1}$, which were suppressed for ease of presentation. We use returns in excess of the Fama-French (1993) three factors and Carhart's (1997) momentum factor over the relevant time window, using betas estimated from 21 to 250 trading days prior to the earnings announcement with a minimum of 3 trading days' data required for parameter estimation. The regression also includes dummies for industry, year-quarter, missing ADV_{t-1} , missing RND_{t-1} , missing $IndADV_{t-1}$ and missing $IndRND_{t-1}$. An event is Post05 if the earnings announcement date occurred during years 2005-onwards, similarly for Post01. Standard errors are reported in parentheses.

***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

If CSR has a positive impact on earnings, then I would expect that it positively contributes to shareholder value. Indeed, I find that Tobin's q, as a market-based measure of shareholder value, is significantly and positively associated to a company's level of CSR throughout the 1992-2009 period, thus supporting the hypothesis (H6) that investors positively incorporated good CSR performance into their assessment of companies' prospects and thus fundamental value. I also find that Tobin's q is significantly higher for CSR+ companies relative to their CSR- counterparts post-2005 (supporting H7); differentials almost doubled compared to 1992-2005, suggesting that investors are getting better at incorporating the full value relevance of CSR information into their assessment of a company's prospects. This evidence of improvement in market efficiency is congruent with our findings that CSR Alpha is disappearing post-2005; it suggests that CSR performance as defined by Statman & Glushkov is becoming more fully priced in the market value of these securities. (Also see Table 14 in Appendix X.)

Table 4: Tobin's Q Regression Results by Period

Period	CSR_{t-1}	GOV_{t-1}	$POST05_t^* CS$ R_{t-1}	$POST05_t$	Cons.	Obs	R^2	Adj R^2	F Stat	P Value
1992-2005†	0.0285 *** (0.0090)	-0.0144 *** (0.0037)			0.0585 (0.0863)	1081	0.5315	0.4976	15.6509	0.0000
2006-2009	0.1006 *** (0.0185)	-0.0168 *** (0.0058)			0.2993 * (0.1715)	390	0.5034	0.4233	6.2877	0.0000
1992-2009†	0.0411 *** (0.0090)	-0.0151 *** (0.0037)	0.0310 * (0.0187)	-0.0076 (0.0244)	0.0864 (0.0745)	1471	0.4570	0.4265	15.0177	0.0000
1992-2009†‡	0.0391 *** (0.0092)	-0.0147 *** (0.0037)	0.0338 * (0.0188)	0.0101 (0.0352)	0.0519 (0.0767)	1401	0.4468	0.4142	13.6914	0.0000

Notes: Reports OLS estimation results of $\text{Log}(Q)_t$ on factors appearing in the table, and $\text{log}(\text{Assets})_t$, $\text{log}(\text{Age})_t$, ROA_t , Leverage_t , Capex_t , GOV_{t-1} , ADV_t , RND_t , IndCapex_t , IndADV_t , IndRND_t , and IndLeverage_t , which were suppressed for ease of presentation. The regression also includes dummies for industry, year-quarter, missing ADV_{t-1} , missing RND_{t-1} , missing IndADV_{t-1} and missing IndRND_{t-1} .
 ***, ** and * are significant at the 1, 5, and 10 percent level, respectively.
 ‡Excluding observation year 2005.
 †Included controls Post01_t and $\text{Post01}_t * \text{GOV}_{t-1}$, which were suppressed for ease of presentation.

As would be expected from the above presented evidence, our analyses on ROA show that CSR is positively associated to operating performance through 1992-2009 (supporting H1; see Table 15 in Appendix 11).

Table 5: ROA Regression Results for 1992-2009

CSR_{t-1}	Capex_t	RND_t	ADV_t	IndCape_t x_t	IndADV_t	IndRND_t D_t	Cons.	Obs	R ²	Adj R ²	F Stat	P Val
0.0328 **	0.0178 ***	-0.0296 ***	0.0056 (0.006)	-0.2007 (0.5678)	-1.3973 ** (0.6005)	0.2798 (0.3711)	-0.2238 ** (0.1044)	143 3	0.295 9	0.257 0	7.604 2	0.000 0

Notes: Reports OLS estimation results of $\text{Log}(Q)_t$ on CSR during 1992-2009, other factors appearing in the table, and $\text{Log}(\text{BME})_{t-1}$, $\text{Log}(\text{Assets})_t$, $\text{Log}(\text{Age})_t$ and GOV_{t-1} , which were suppressed for ease of presentation. The regression also includes dummies for industry, year-quarter, missing ADV_{t-1} , missing RND_{t-1} , missing IndADV_{t-1} and missing IndRND_{t-1} .
 ***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

Finding that CSR is positively associated to ROA and Tobin's q is congruent with results of many other empirical studies (see sections II and III). However, finding that investors have historically been surprised by CSR's contribution to companies' earnings prospects is unique to this study, and – I believe – an important contribution to the field of both capital markets and strategic management research (discussed in the next section). By looking at measures of stock market efficiency, not only does the tale of CSR's Alpha suggest that CSR *causes* (Joshua et al., 1996) superior operating performance by its association to analyst forecast errors and earnings announcement surprises (supporting H10), but it also suggests that CSR causes superior shareholder value: I find that investors were differentially less surprised at the actual earnings performance of CSR+ companies if they already more fully incorporated the value relevance of CSR information in their assessment of a company's prospects as reflected by higher Tobin's q values (see Tables 2 and 3). In line with our hypotheses (H8 and H9), the value and significance of the terms $Q_{t-1} * \text{CSR}_{t-1}$ and Q_{t-1} in our analyses of analyst forecast errors and earnings announcement surprise support the hypothesis (H11) that CSR causes superior shareholder value.

Uncovering these relationships brings clarity to a 40-year-old debate. It disputes the anti-thesis that CSR leadership is associated to superior operating performance and shareholder value because of a reverse causality relationship based on the slack resources hypothesis (Preston & O'Bannon, 1997; Margolis et al., 2007; McGuire et al. 1988, 1990; Orlitzky et al., 2003; Preston & O'Bannon, 1997; Seifert et al., 2004; Ullmann 1985; Waddock & Graves, 1997).

Taken together, findings suggest that CSR *caused* stock returns during period 1992-2005 but that CSR's Alpha seems to be disappearing following recent industry developments in the area of ESG incorporation. Investors would now be more fully incorporating the value relevance of

CSR information into their expectations of companies' prospects and thus valuation, becoming less surprised by the actual operating performance of CSR leaders post-2005. It is possible that CSR is now priced in the market value of S&P 500 constituents to the point that CSR Alpha might be much more difficult to achieve via best-in-class methods, but suggesting that CSR leaders now benefit from a lower cost of capital.

X. Discussion and Implications

Alpha's Tale begins with the Hypothesis that CSR is a source of competitive advantage that creates economic value for leading companies. It is the central premise behind the tale of CSR's Alpha; that CSR – rather than risk or other, period specific drivers – caused stock returns because investors underestimated its value relevance. 40 years ago, Milton Friedman anchored the argument that CSR is detrimental to shareholder value. Around the same time, Milton Moskowitz began offering investors socially responsible investment picks for those wanting to invest according to their ethics/ values in line the view that investing according to CSR criteria is the principal domain of those investors who don't mind sacrificing some investment profits to feel better about their portfolio. Thus began CSR's representation in management and capital markets practice and research: CSR conflicted with Friedman's profit maximand and responsible investment by CSR criteria was detrimental to portfolio risk and return. This belief became entrenched in the collective representation of both communities.

Significant shifts in conventional beliefs began around the turn of the century. Stakeholder theory (Donaldson & Preston, 1995; Freeman, 1984), resource dependence theory (Pfeffer & Salancik, 1978; Ulrich & Barney, 1984) and institutional theory (DiMaggio & Powell, 1983; Meyer & Rowan, 1977) lent support to the claim that companies were dependent upon a wide variety of stakeholders to ensure the long-term sustainability of their business. First, theories on strategic philanthropy and, later, other forms of CSR were put forward by academics as corporate activities that can potentially create shareholder value. Empirical evidence that ensued both supported and refuted the Hypothesis of value creation, leading to a fierce debate.

As the body of work centering on the debate got bigger, it became clearer that CSR is a source of competitive advantage (Baron, 2000; Porter & Kramer, 2006). Several studies reporting that investment strategies based on CSR information produced Alpha (Derwall et al., 2005; Edmans, 2010; Kempf & Osthoff, 2007; Statman & Glushkov, 2009) further supported the hypothesis, but also hinted that investors had been giving inadequate attention to such value relevant information.

Our findings suggests that the fundamental value of CSR as it relates to companies' operating performance and earnings prospects is so complex (Branco & Rodrigues, 2006; Hillman & Keim, 2001) that a trading strategy based on CSR information has generated more consistent investment profits year-after-year in period 1992-2005 than the trading strategy based on governance information as put forward by Gompers et al. (2003) (Bebchuk et al., 2010; Core et al., 2006). Market inefficiency of the sort proposes that investors act with bounded rationality (Simon, 1957; Radner & Rothschild, 1975), but there are, most likely, other forces at play such as agency problems (Guyatt, 2005) and institutional pressures that exist at various levels of the market system (Badrinath et al. 1989; Coates & Davis, 1995; Del Guercio, 1996; Scharfstein &

Stein, 1990), which over-emphasizes short-term results (Eccles et al., 2001; Graves & Waddock, 1990; Jacobs, 1991; Porter, 1992), and discourages collaborative action (Ouchi, 1980; Oye, 1986; Williamson, 1985). All these forces would have contributed to and reinforced the inefficiency.

Thus our findings suggest that there is an important place for the study of the effects of investor psychology (Shleifer, 1999; Shefrin, 2002; Shiller, 2000) and institutional pressures (DiMaggio & Powell, 1983; Meyer & Rowan, 1977; Weber, 1978; Zucker, 1977) on capital markets, and ample room for policy makers and other institutional players such as the UN PRI to improve efficiency. These are promising avenues for research and a relevant area for new regulation.

Mispricing in capital markets is known to affect (Froot et al., 1992b; Jensen, 2004; Rappaport, 2005) and reinforce (Bandura, 1977) bad management decisions. For example mispricing towards value contributing R&D activities (Chan et al. 2001; Lev and Sougiannis, 1996) has been shown to discourage allocation to the activity by companies (Graves & Waddock, 1990). Similarly, CSR mispricing, might have exacerbated myopic corporate behaviour (Drucker 1986; Eccles et al., 2001; Jacobs, 1991; Porter, 1992) towards CSR.²⁸

By improving investors' attention to and thus treatment of CSR information, initiatives such as the UN PRI, which have helped shape the institutional environment of the investment industry, have, by improving efficiency, also encouraged more responsible corporate behaviour and thus more sustainable portfolios. CSR leaders now benefit from lower costs of capital. However, although improved market efficiency is desirable, findings lay doubt on the claim that investors will be able to continue using best-in-class trading strategies to obtain CSR Alpha. That is not to say that CSR investment profits have disappeared completely, but rather that they are less likely to exist – or more difficult to obtain – via best-in-class trading strategies such as Statman & Glushkov's.

PRI signatories, which number over 900 and represent over 20 Trillion USD in assets under management, have pledged to incorporate CSR into their investment processes and ownership practices: In light of the eventual disappearance of CSR Alpha, responsible investors will shift their strategies away from security election, towards other novel approaches to strategic asset allocation, impact investing and shareholder engagement in order to capitalize on these promising forms of economic value creation.

²⁸ Konar & Cohen's (1997) finding that companies that experience the most economically significant stock price declines were subsequently the most improved in terms of environmental management, supports this disciplinary function of capital markets.

The most important asset owners today have a long-term perspective on capital markets (Davis, 2002; Monks, 2001),²⁹ and therefore value long-term investment in R&D as well as other capital expenditures like CSR (Jarrell et al., 1985; Levis, 1989; Marsh, 1990; McConnell & Muscarell, 1985). The Freshfields Report and its follow-up report (UNEP FI, 2009) do a good job at outlining many reasons why those investors would be legally required to consider the CSR of current and prospective investee companies in their mainstream investment processes and ownership practices. Together with these reports, the works of Monks (2001) and Hawley & Williams (2000) on the Universal Owners Hypothesis provide a good theoretical argument for such considerations; an argument that is grounded in their ownership of a cross-section of the economy, the increasing trend towards passive investing, and the long-term nature of their liabilities. Peter Drucker (1976) proclaimed that these (then) emerging trends would lead to ‘pension fund socialism’.

These Universal Owners represent similar fiduciary interests. By acting together, they can exercise substantial power over companies with the goal of persuading or requiring the adoption of CSR where such an activity improves shareholder value, helping to resolve the myopic corporate behaviour by corporations (Black, 1992) and enhancing portfolio value (Thamotheran & Wildsmith, 2007). As such, they can contribute to company specific financial performance as well as encourage industry-level best-practices that encourage more sustainable systematic behaviour (Hebb, 2008). Brad Barber (2007) found that CalPERS, one of the largest such investors, created over \$89.5 billion in economic benefits during 1992-2005 by proactively engaging companies to raise their corporate governance standards. The potential for similar engagement from these investors to encourage companies to increase improve CSR seems promising. The strategies and the impact of shareholder engagement on CSR promises to be a fruitful for research.

A relatively unexplored and controversial implication of the Universal Owners Hypothesis concerns encouraging companies to engage in the type of CSR that would have a net negative financial impact on its fundamental value, but a net positive impact on a Universal Owners’ portfolio. Although the institutional environment might never permit this type of CSR, it is conceivable that Universal Owners might eventually turn to a new form of investing, called ‘impact investing’ (Monitor Institute, 2009), with the wider ambition to invest some of their profits from normal investment activities into non-profit social and environmental organisations that could improve various companies’ operating context and thus enhance their prospects and thereby improve portfolio value. It is also foreseeable that such investors might, through shareholder engagement, promote corporate policies that discourage companies from acting in socially irresponsible ways – activities that would have otherwise been net positive on that company’s value but net negative on Universal Owners’ portfolios. Universal Owners are also expected to increase their engagement with public policy makers to the same end; that is, to use

²⁹ These include pension funds and insurance companies.

regulatory mechanisms that encourage (discourage) positive (negative) social externalities from companies' regular business activities. This promises to be a fruitful avenue for future research.

CSR is evolving along the same lines as responsible investing: In a 2011 HBR article, Porter & Kramer expanded upon the concept of 'Creating Shared Value', an idea first introduced in their 2006 HBR article. In a recent interview by Adi Ignatius of the Harvard Business Review (HBR), Porter (2011) explained his new perspective on CSR in the following way:³⁰

"We once thought: if business just increases its profit, what's good for business is then good for society... we need to think differently: What's good for society is actually good for business... The concept of shared value says that actually creating societal benefit is really a powerful way to create economic value for the company, and that we miss by-and-large many of these opportunities to create profit – let's call it: 'the right way' – profit that comes from meeting societal needs."

Although the last 40 years of CSR research has focused on identifying areas where CSR accorded with Milton Friedman's profit maximand, It is expected that the next 40 years of CSR/responsible investment research will focus on identifying those areas for shared value creation, and the measurement of the value created, not only at the actual company level, but also at the societal-level.

This process would be driven first by the identification of societal issues to be potentially resolved by business, by the crafting of prospective solutions, and then by the measurement of shareholder value that would be created by solving these issues. It is the difference between a perspective that pits the goals of business against those of wider society and one that recognizes their interdependence. Another term for this concept is 'creative capitalism'. In this new understanding of CSR, there is much room for novel corporate partnerships and alliances with companies in different industries and different geographic areas, and with a myriad of other organisations including non-profits, universities, governments, local and global communities, foundations, investors, etc. It is a promising avenue for future research.

Taking account of our findings in the context of these trends suggests that many opportunities that had historically been latent in our market system will become increasingly thought of as important solutions to make our companies more accountable, more socially responsible and more sustainable, thereby enhancing companies' and investments' returns to the economy and society at large (Lydenberg, 2009). It suggests a new role for non-governmental organisations and other interest groups with investors and asset owners playing an increasingly important political and economic role.

³⁰ Available at: <http://hbr.org/2011/01/the-big-idea-creating-shared-value/ar/1>

Lastly, what are the limits of CSR and responsible investing as important virtues of the market system (see Entine, 2003; Karnani, 2007; Richardson, 2009; Richardson & Cragg, 2010; Vogel, 2006)? And what is to say of government's role in society: how should governments address the challenges and opportunities discussed in this section (eg. McKinsey & Company, 2009; Rhodes, 1996)? These questions will continue to be fruitful avenues for research.

XI. Conclusion

This paper sought to uncover the causality relationship at play in CSR's Alpha. Restricting our analyses to S&P 500 constituents, I found that an equally-weighted, long-only portfolio of good CSR companies produced a significant 4.3 percent p.a. risk-adjusted investment profit in years 1992-2005. Alpha's Tale suggests that this investment profit was due to CSR mispricing by showing that investors misevaluated the importance of CSR performance for future operating cash flows; analyst forecast errors were significantly larger for CSR leaders (CSR+) than they were for laggards (CSR-), indicating that investors failed to forecast the annual earnings potential generated by good CSR performance at beginning of fiscal year. Investors were consequently significantly more surprised at the quarterly earnings performance of CSR leaders and, upon being surprised, re-evaluated their assessment of these companies' prospects, resulting in an increase of the market valuation of their market valuation.

Tobin's q is a market-based measure of companies' fundamental shareholder value. Despite CSR mispricing, CSR leadership was significantly associated with higher Tobin's q in 1992-2005. However, I found that investors were significantly less (more) surprised at the earnings performance of CSR leaders if they had more (less) fully incorporated the value relevance of CSR information into their assessment of these companies' earnings prospects (as defined by a higher Tobin's q); they were less surprised at the earnings performance of CSR+ companies that had already been attributed a higher Tobin's q value by the market. These findings suggest that CSR was not fully priced in during 1992-2005, and that CSR not only causes (Joshua et al., 1996) superior operating performance and shareholder value.

CSR mispricing seems to have been prevalent throughout 1992-2005: Various equities trading strategies based on CSR information would have significantly outperformed during this period (Derwall et al., 2005; Edmans, 2010; Kempf & Osthoff, 2007; Statman & Glushkov, 2009), contradicting the assumption that stock markets are semi-strong form efficient (Fama, 1970). Relying on agency theory (Eisenhardt, 1988, 1989; Perrow, 1968), bounded rationality (Simon, 1957; Radner & Rothschild, 1975) and unfavourable institutional pressures (Badrinath et al. 1989; Coates & Davis, 1995; Del Guercio, 1996; Scharfstein & Stein, 1990), I supplied various explanations for this inefficiency (section V) and outlined how developments in the institutional environment of the investment industry improved investors' attention to and treatment of CSR information post-2005 (section VI). These developments include a notion that ESG performance can contribute to investment outperformance, that the incorporation of ESG information into investment processes may be required by fiduciary duty, and the impact of institutional isomorphism that resulted from various initiatives such as the Enhanced Analytics Initiative and the United Nations Principles for Responsible Investment. The post-2005 breakpoint has also been shown to correspond to a change in investor attitude towards ESG, recognizing that crafting

investment strategies according to such information is not solely restricted to the domain of ethics-/ values-based investors.

As a result of the market's improved pricing of CSR information, the trading strategy based on an equally-weighted, long-only portfolio of good CSR companies yielded investment profits that were significantly lower post-2005 than they had been in 1992-2005. Investment outperformance might now be difficult to achieve via security selection by CSR criteria; CSR may be fully priced in the market value of these companies. Our finding that Tobin's q was significantly larger for CSR+ companies post-2005 than they had been 1992-2005 supports our hypothesis of improved market efficiency towards CSR information. Analyst forecast errors and earnings announcement surprises also point to an improvement in CSR pricing post-2005.

Alpha's Tale has various implications for investment/ capital markets and strategic management researchers and practitioners. The disappearance of CSR Alpha implies that responsible investment strategies will move away from security selection and towards new approaches to strategic asset allocation (such as impact investing) and a more important focus on corporate engagement. Section II presented a comprehensive theory on the mechanisms by which CSR delivers competitive advantage by enhancing risk management practices, supporting the development of valuable capabilities, and improving the management of stakeholder interests. Knowing that increasing CSR expenditures *generates* shareholder value, investors can use our findings to encourage companies to adopt CSR policies that enhance shareholder value.

Our findings applaud initiatives such as the UN PRI which, by improving investors' attention to and treatment of CSR information, has corrected the market's signal to companies and thereby encouraged more sustainable corporate behaviour and better returns to the economy and society at large. Because companies that engage in value contributing CSR post-2005 were better rewarded by the market than they were 1992-2005, I foresee that companies will increasingly think of CSR as a legitimate corporate activity. I also foresee that CSR will become increasingly recognized for its ability to create shared value both at the societal and corporate level, in a demarcation away from Milton Friedman's strict profit maximand and towards social and economic impact.

XII. Reference list

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XIII: Appendixes

Appendix I

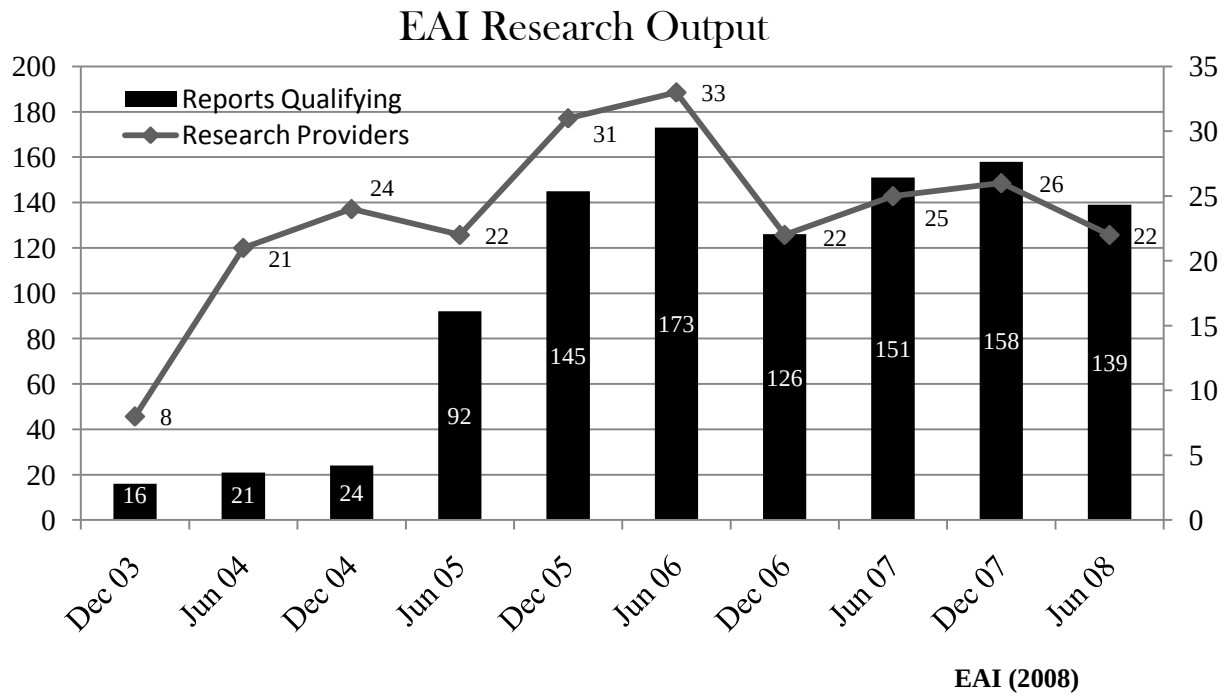
KLD Issues by Characteristic

	Strengths	Concerns
1. Community	Charitable Giving Indigenous Peoples Relations Innovative Giving Non-US Charitable Giving Support for Education Support for Housing Volunteer Programs Other Strength	Indigenous Peoples Relations Investment Controversies Negative Economic Impact Tax Disputes Other Concern
2. Diversity	Board of Directors CEO Employment of the Disabled Gay & Lesbian Policies Promotion Women & Minority Contracting Work/Life Benefits Other Strength	Non-Representation Other Concern
3. Employee Relations	Cash Profit Sharing Employee Involvement Health and Safety Strength No-Layoff Policy Retirement Benefits Strength Union Relations Other Strength	Health and Safety Concern Retirement Benefits Concern Union Relations Workforce Reductions Other Concern
4. Environment	Beneficial Products and Services Clean Energy Communications Pollution Prevention Property, Plant, and Equipment Recycling Other Strength	Agricultural Chemicals Climate Change Hazardous Waste Ozone Depleting Chemicals Regulatory Problems Substantial Emissions Other Concern
5. Human Rights	Indigenous Peoples Relations Strength Labor Rights Strength Positive Record in South Africa Other Strength	Burma Concern Indigenous Peoples Relations Concern Labor Rights Concern Mexico

		Northern Ireland South Africa Other Concern
6. Product	Quality R&D/Innovation Other Strength	Antitrust Marketing/Contracting Concern Product Safety Other Concern

Appendix II

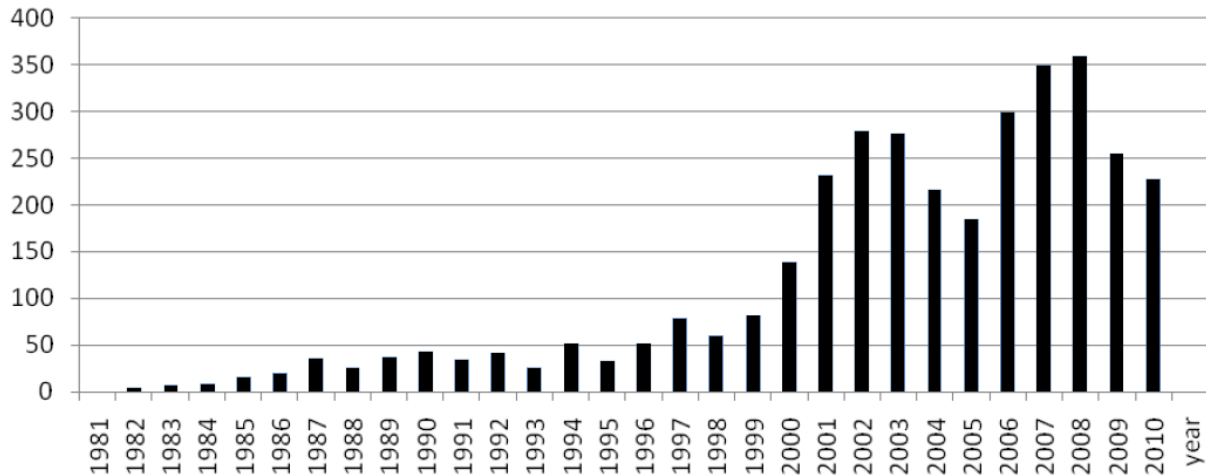
The emergence of ESG sell-side research



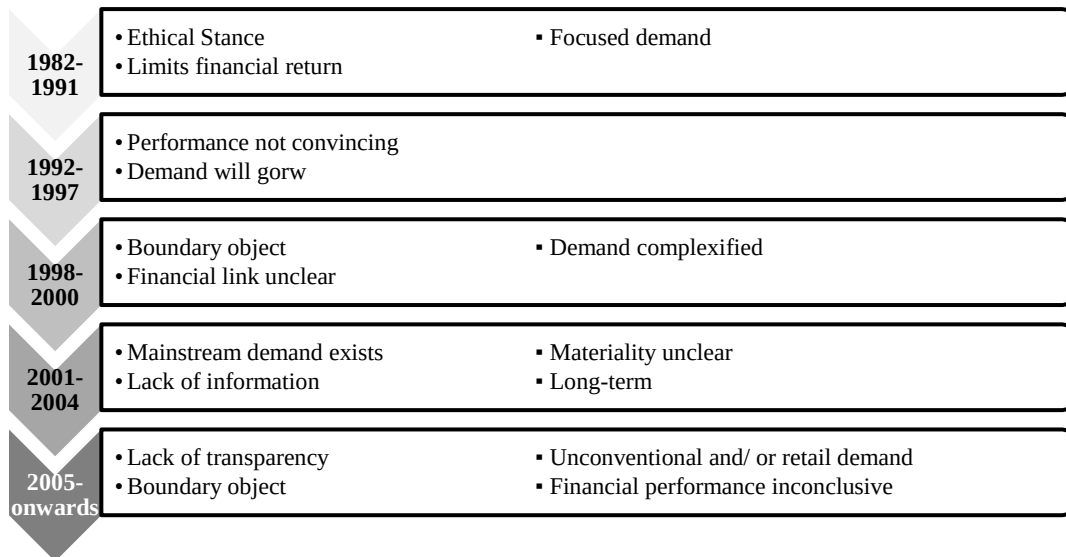
Appendix III

Select figures and tables from Dumas & Louche (2011)

NUMBER OF RI ATRICLES/ YEAR



SALIENT REPRESENTATIONS PER BRACKET



QUOTES TAKEN FROM PERIOD 2005-ONWARDS (ESG YEARS)

Quotes			
1. Financial return	P5Q2.1. "It's no good our saying, 'we don't perform so well but count on us to save the world'. Source: 20060407FT2.	20091122FTP5Q2.2. "Witness the conflicting results from analyses to determine whether investors sacrifice returns for ethical investing. According to Morningstar, US domestic equity SRI funds had an average annualised return of 8.53 per cent over the past three years, while the Standard & Poor's 500 Index posted 10.32 per cent. By stretching the period to 10 years, which includes the dotcom boom, this gap narrows, largely because of SRI funds' technology holdings a 7.21 per cent gain on average each year versus 7.93 per cent for the S&P 500. A Wharton School study over a longer period equally shows SRI funds underperforming a broader universe of funds. Other research, though, finds (...)" Source: 20070305FT	
2. Information availability	availabilityP5Q1.1. "But data are often too vague, lacking order and integrity, and there is no agreement internationally over what a sustainability report should comprise." Source: 20101004FT.2	P5Q12. "Another complication is this: how does one independently confirm proprietary analysis? More work is needed to develop objective, comprehensive and verifiable processes that enable investors to compare companies' ESG performances." Source: 20070305FT	P5Q1.3. "He praises the Dollars 380bn Norway Government Pension Fund, which provides full transparency and has a broadly diversified portfolio, close-to-index market exposures, and self-imposed caps on maximum positions in individual companies." Source: 20080303FT
	P5Q1.4. "There are, however, gaps in the range. Value stocks, often heavy industries which are traditionally unkind to the environment, are not well represented among good SRI performers. Nor are global or overseas funds, partly thanks to a lack of good information on which to base screening decisions." Source: 20060407FTP5Q1.5.	P5Q1.5. "Screening is also hampered by the difficulties inherent in quantifying environmental, social and governance (ESG) issues. Data points may not be readily obtainable. Op-aque corporate disclosures may obstruct data gathering." Source: 20070305FT>	P5Q1.6. "As active owners, investors should work with investee companies to ensure comprehensive and systematic disclosure of the information they need in order to make responsible investment decisions. Ensuring the disclosure of information on ESG and other issues will enhance investors' understanding of their underlying investments and avoid a repeat of recent mistakes." Source: 20090302FT.2>
3. Demand for SRI	P5Q3.1. "Ultimately, though, he thinks the strength of demand from US investors will create new investment channels in the SRI market. After all, research group Celent predicts that SRI assets under management in the US will reach Dollars 3,000bn by 2011." Source: 20071001FT	P5Q3.2. David Anderson, chief executive of Co-operative Financial Services, said it vindicated the bank's cautious lending policies and ethical stance. "People are attracted to our model. There is more demand than we can fulfill. We will lend more in 2009 than in 2008," he said, although growth would probably slow. Source: 20090403FT	
4. Boundary object	P5Q4.1. First comes greater standardisation of SRI performance measurement. The Social Investment Forum concluded: "With different definitions of SRI, market factors, cultural concerns and methodologies for collecting data, it is difficult to make controlled comparisons on a global scale." Source: 20070305FT	P5Q4.2. When people say they want to invest ethically, we ask them what they mean," he explains. Source: 20090411FT	P5Q4.3. Sarasin's definition of sustainable investment is by its own admission somewhat subjective. (...) The trouble with nuclear power is that the question of how and where you store the nuclear waste remains unresolved. As long as it's unresolved it's unsustainable." Source: 20091122FTP5Q2.2. " "

Appendix IV

TABLE6: CSR- LONG-ONLY MULTIFACTOR REGRESSION RESULTS BY PERIOD

<i>Period</i>	<i>α</i>	<i>$Rm - R_{ft}$</i>	<i>SMB</i>	<i>HML</i>	<i>MOM</i>
1992- 1995	0.02%	1.0263 ***	0.1610 ***	0.2337 ***	-0.1021
1993- 1996	0.00%	1.0603 ***	0.1915 **	0.0660	-0.0975
1994- 1997	0.80%	1.0372 ***	0.0949	-0.0230	-0.1859 **
1995- 1998	-0.01%	0.9352 ***	0.0905	0.1076	-0.4017 ***
1996- 1999	-0.45%	0.9535 ***	0.0481	0.1808	-0.4382 ***
1997- 2000	-1.90%	1.0082 ***	0.0832	0.5530 ***	-0.3479 ***
1998- 2001	-2.45%	0.9701 ***	0.0782	0.7060 ***	-0.1737 **
1999- 2002	2.55%	1.0244 ***	0.0507	0.7005 ***	-0.1829 ***
2000- 2003	2.03%	1.0194 ***	0.0924	0.7464 ***	-0.2040 ***
2001- 2004	5.57%	1.0641 ***	0.1662 **	0.6596 ***	-0.1264
2002- 2005	2.24%	1.0393 ***	0.1774 **	0.5325 ***	-0.2711 ***
2003- 2006	-1.19%	1.1810 ***	-0.0144	0.3140 *	-0.18691 **
2004- 2007	0.70%	1.0005 ***	0.0380	0.3677 **	-0.1378
2005- 2008	-1.92%	1.1979 ***	-0.0123	0.2457	-0.2006 **
2006- 2009	-1.64%	1.1249 ***	0.1567	-0.2482 *	-0.2598 ***
2007- 2010	-0.18%	1.1487 ***	0.2579 **	-0.2750 **	-0.2389 ***

Notes: Reports a sub-period breakdown of monthly regression results of an equal-weighted long-only portfolio of CSR- companies. For each year, average alpha is estimated using 12 months of data corresponding to the calendar year. α is an annualized percentage. White (1980) robust standard errors are used throughout and are reported in parentheses. The Fama-French (1993) three-factor model enhanced with the Carhart (1997) momentum factor was used as a benchmark.

***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

TABLE 7: CSR- LONG ONLY MULTIFACTOR REGRESSION RESULTS

<i>Period</i>	<i>α</i>	<i>$Rm - R_{ft}$</i>	<i>SMB</i>	<i>HML</i>	<i>MOM</i>	<i>Post05</i>	<i>R²</i>
1992- 2005	0.08% (0.001759)	1.0025 *** (0.0423)	0.0843 ** (0.0425)	0.5741 *** (0.0613)	-0.2132 *** (0.0554)		0.8268
2005- 2010	0.01% (0.002346)	1.1307 *** (0.0764)	0.06423 (0.0984)	-0.1013 (0.1262)	-0.2118 *** (0.0392)		0.9218
1992-2010	1.22% (0.00177)	0.9803 *** (0.0526)	0.0304 (0.0420)	0.4284 *** (0.0770)	-0.1990 *** (0.0437)	-2.45% (0.003296)	0.8505
1992-2009‡	0.99% (0.0019)	0.9681 *** (0.0607)	0.0372 (0.0433)	0.4327 *** (0.0856)	-0.2065 *** (0.0472)	-3.85% (0.0037)	0.8466

Notes: Reports a breakdown of monthly regression results of an equal-weighted long-only portfolio of CSR- companies by sub-period. For each year, average alpha is estimated using 12 months of data corresponding to the calendar year. α and Post05 are annualized percentages. White (1980) robust standard errors are used throughout and are reported in parentheses. The Fama-French (1993) three-factor model enhanced with the Carhart (1997) momentum factor was used as a benchmark.

***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

‡Excluding year 2005.

Appendix V

TABLE 8: CSR+ LONG-ONLY MULTIFACTOR REGRESSION RESULTS BY PERIOD

<i>Period</i>	<i>α</i>	<i>$R_m - R_{ft}$</i>	<i>SMB</i>	<i>HML</i>	<i>MOM</i>
1992- 1995	3.73%	1.0211 ***	-0.0312	-0.1057	0.0300
1993- 1996	2.52%	0.9589 ***	-0.2383 ***	-0.1832 **	0.0406
1994- 1997	0.58%	0.9520 ***	-0.1579 ***	-0.1614 *	0.0128
1995- 1998	0.08%	0.9847 ***	-0.1874 **	-0.0839	-0.1061
1996- 1999	0.94%	0.9995 ***	-0.1743 ***	0.0347	-0.2375 ***
1997- 2000	6.69% *	1.0376 ***	-0.0581	0.3100 **	-0.1420 *
1998- 2001	9.61% ***	1.0627 ***	0.0055	0.4317 ***	-0.0995 **
1999- 2002	12.01% ***	1.0514 ***	-0.0288	0.3771 ***	-0.1266 ***
2000- 2003	11.50% **	1.0286 ***	-0.0426	0.3310 ***	-0.1090 **
2001- 2004	5.30%	1.0686 ***	0.0898	0.2011	-0.0186
2002- 2005	5.35% *	1.0792 ***	0.1035	-0.1735	-0.0518
2003- 2006	0.36%	1.0529 ***	0.1945	-0.1860 *	-0.0777
2004- 2007	-0.50%	1.0215 ***	0.2783***	-0.1813***	-0.1737***
2005- 2008	-0.68%	1.0239 ***	0.2459 ***	-0.1500 **	-0.1732 ***
2006- 2009	-0.82%	1.0142 ***	0.3078***	-0.0578	-0.1687***
2007- 2010	0.93%	1.0350 ***	0.3404 ***	-0.0145	-0.1368 ***

Notes: Reports a sub-period breakdown of monthly regression results of an equal-weighted long-only portfolio of CSR+ companies. For each year, average alpha is estimated using 12 months of data corresponding to the calendar year. α is an annualized percentage. White (1980) robust standard errors are used throughout and are reported in parentheses. The Fama-French (1993) three-factor model enhanced with the Carhart (1997) momentum factor was used as a benchmark.

***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

Appendix VI

TABLE 9: EARNINGS ANNOUNCEMENT SURPRISES BY PERIOD

PANEL A: PERIOD 1992-2005†									
<i>Window</i>	<i>CSR_{t-1}</i>	<i>Q_{t-1}*CSR_{t-1}</i>	<i>Q_{t-1}</i>	<i>Cons.</i>	<i>Obs.</i>	<i>R²</i>	<i>Adj. R²</i>	<i>F Stat</i>	<i>P-Value</i>
-1:+1	0.003891 * (0.002007)	-0.017346 (0.010875)	0.01228 (0.012913)	-0.054825 * (0.029025)	4188	0.026969	-0.00002	0.999268	0.485194
-3:+1	0.003909 * (0.002216)	-0.020331 * (0.012007)	0.015646 (0.014258)	-0.045864 (0.032046)	4188	0.031382	0.004516	1.168075	0.110735
-5:+1	0.004501 * (0.002386)	-0.016556 (0.012933)	0.012193 (0.015357)	-0.035573 (0.034518)	4188	0.041212	0.014618	1.549677	0.00021
-10:+1	0.005009 * (0.002892)	0.006418 (0.015672)	-0.005403 (0.01861)	-0.044455 (0.041829)	4188	0.065809	0.039898	2.53976	0
-20:+1	0.006024 (0.003662)	0.043595 ** (0.019845)	-0.044329 * (0.023565)	-0.047158 (0.052966)	4188	0.088016	0.06272	3.479504	0
PANEL B: PERIOD 2006-2010									
<i>Window</i>	<i>CSR_{t-1}</i>	<i>Q_{t-1}*CSR_{t-1}</i>	<i>Q_{t-1}</i>	<i>Cons.</i>	<i>Obs.</i>	<i>R²</i>	<i>Adj. R²</i>	<i>F Stat</i>	<i>P-Value</i>
-1:+1	0.003033 (0.004742)	-0.005518 (0.027747)	0.015556 (0.03101)	-0.012251 (0.015871)	1916	0.033793	-0.00124	0.964682	0.559041
-3:+1	0.000833 (0.004942)	-0.018541 (0.028915)	0.027718 (0.032315)	-0.01366 (0.016539)	1916	0.034016	-0.00101	0.971267	0.543909
-5:+1	0.000819 (0.005342)	-0.034865 (0.031254)	0.034372 (0.03493)	-0.009365 (0.017877)	1916	0.041501	0.006751	1.194256	0.137563
-10:+1	0.001885 (0.006417)	-0.048481 (0.037545)	0.043231 (0.041961)	-0.013119 (0.021475)	1916	0.04648	0.01191	1.344506	0.034659
-20:+1	0.001317 (0.007554)	-0.020405 (0.044196)	0.046352 (0.049394)	-0.035764 (0.025279)	1916	0.070973	0.037291	2.107135	0.000001

Notes: Reports the relationship between earnings announcement returns (windows ranging -1, -3, -5, -10 and -20 to +1 trading days around quarterly earnings announcement dates), and CSR performance using OLS regression of the announcement returns in a particular window on the terms listed in the table and Gov_{t-1} , ADV_{t-1} , RND_{t-1} , $IndADV_{t-1}$, $IndRND_{t-1}$, $Q_{t-1}*Gov_{t-1}$, $Q_{t-1}*ADV_{t-1}$ and $Q_{t-1}*RND_{t-1}$, which were suppressed for ease of presentation. We use returns in excess of the Fama-French (1993) three factors and Carhart's (1997) momentum factor over the relevant time window, using betas estimated from 21 to 250 trading days prior to the earnings announcement with a minimum of 3 trading days' data required for parameter estimation. The regression also includes dummies for industry, year-quarter, missing ADV_{t-1} , missing RND_{t-1} , missing $IndADV_{t-1}$ and missing $IndRND_{t-1}$. Standard errors are reported in parentheses. ***, ** and * are significant at the 1, 5, and 10 percent level, respectively. †Including controls $Post01_t$, $Post01_t*Gov_{t-1}$, which were suppressed for ease of presentation

TABLE 10: ANALYST FORECAST ERRORS BY PERIOD

PANEL A: PERIOD 1992-2006†									
<i>Scale</i>	<i>CSR_{t-1}</i>	<i>Q_{t-1}*CSR_{t-1}</i>	<i>Q_{t-1}</i>	<i>Cons.</i>	<i>Obs.</i>	<i>R²</i>	<i>Adj R²</i>	<i>F Stat</i>	<i>P-value</i>
$Assets_{t-1}$	0.0097 *** (0.0022)	-0.030474 ** (0.0123)	0.0144 (0.0146)	0.0231 (0.0217)	1167	0.1643	0.1093	2.9870	0.0000
PPS_{t-1}	0.0111 *** (0.002726)	-0.045595 *** (0.0147)	0.0575 *** (0.0175)	0.0125 (0.0260)	1167	0.1265	0.0690	2.2007	0.0000
PANEL B: PERIOD 2007-2009									
<i>Scale</i>	<i>CSR_{t-1}</i>	<i>Q_{t-1}*CSR_{t-1}</i>	<i>Q_{t-1}</i>	<i>Cons.</i>	<i>Obs.</i>	<i>R²</i>	<i>Adj R²</i>	<i>F Stat</i>	<i>P-value</i>
$Assets_{t-1}$	0.0056 (0.0064)	0.0266 (0.0365)	-0.0223 (0.0422)	-0.0404 * (0.0243)	294	0.1967	0.0314	1.1901	0.1966
PPS_{t-1}	-0.0047 (0.0165)	0.0949 (0.0946)	0.0278 (0.1092)	-0.1678 *** (0.0627)	294	0.4749	0.3669	4.3956	0.0000

Notes: Reports the relationship between analyst forecast errors (scaled by $Assets_{t-1}$ and PPS_{t-1}) and CSR performance using OLS regression of the announcement returns on the terms listed in the table, and Gov_{t-1} , ADV_{t-1} , RND_{t-1} , $IndADV_{t-1}$, $IndRND_{t-1}$, $Q_{t-1}*Gov_{t-1}$, $Q_{t-1}*ADV_{t-1}$ and $Q_{t-1}*RND_{t-1}$, which were suppressed for ease of presentation. The regression also includes dummies for industry, year, missing ADV_{t-1} , missing RND_{t-1} , missing $IndADV_{t-1}$ and missing $IndRND_{t-1}$. ***, ** and * are significant at the 1, 5, and 10 percent level, respectively. †Including controls $Post01_t$, $Post01_t*Gov_{t-1}$, which were suppressed for ease of presentation

Appendix VII

TABLE 11: FORECAST ERRORS BY PERIOD, SCALED BY PPS (T-1)

<i>Period</i>	CSR_{t-1}	$Q_{t-1} * CSR_{t-1}$	Q_{t-1}	Gov_{t-1}	ADV_{t-1}	RND_{t-1}	$IndADV_{t-1}$	$IndRND_{t-1}$	$Q_{t-1} * Gov_{t-1}$	$Q_{t-1} * ADV_{t-1}$	$Q_{t-1} * RND_{t-1}$	<i>Cons.</i>
1992-1995	0.0068	0.0502	0.0389	-0.0021	0.0045	-0.0007	-0.4909	-0.3523	-0.0367 ***	-0.0196	0.0249	0.0383
1993-1996	0.0075	0.0398	0.0022	-0.00295 *	0.0039	0.0017	-0.3246	0.1408	-0.0198 *	-0.0261	0.0440 **	0.0163
1994-1997	0.0100 **	0.0276	-0.0287	-0.0037 **	0.0008	0.0022	-0.2100	0.1943	-0.0094	-0.0171	0.0328 *	0.0150
1995-1998	0.0135 ***	-0.0249	0.0168	-0.0040 **	0.0026	-0.0216 ***	-0.3058	-0.2679	-0.0083	-0.0174	0.0740 ***	0.0342
1996-1999	0.01305 **	-0.0890 ***	0.0432	-0.0034 *	0.0022	-0.0358 ***	-0.3201	-0.4812	-0.0044	-0.0331 *	0.1160 ***	0.0584
1997-2000	0.0076	-0.1478 ***	0.0390	-0.0032 **	-0.0023	-0.0410 ***	0.3262	-1.8220 ***	-0.0010	-0.0103	0.133087 ***	0.0250
1998-2001	0.0056	-0.1039 ***	0.0307	-0.0015	0.0011	-0.0394 ***	0.1022	-0.7612 *	-0.0048	-0.0047	0.1122 ***	0.0346
1999-2002	0.0073	-0.1023 ***	0.0090	-0.0019	-0.0045	-0.0163 ***	0.2219	-0.6123 ***	-0.0005	0.0184	0.0815 ***	0.0044
2000-2003	0.0083 *	-0.0081	0.0477 *	-0.0002	0.0000	0.0012	-0.0443	-0.1513	-0.0078	0.0119	-0.0082	0.0064
2001-2004	0.0094 **	-0.0195	0.0661 **	-0.0008	0.0005	0.0003	-0.0149	-0.1068	-0.0096	-0.0005	-0.0067	0.0054
2002-2005	0.0079 *	-0.0330	0.1033 ***	0.0006	0.0028	0.0003	0.3661	0.1316	-0.0135	-0.0076	-0.0081	-0.0271
2003-2006	0.0128 *	-0.0365	0.1043 **	0.0023	-0.0027	0.0036	0.4282	0.0657	-0.0124	0.0087	-0.022497 *	-0.0347
2004-2007	0.0064	-0.0806	0.1256 **	0.0000	-0.0040 *	-0.0006	0.1964	0.1064	-0.0107	0.0108	-0.0087	-0.0047
2005-2008	-0.0005	-0.0608	0.1464 *	0.0093 **	-0.0043	0.0034	0.6024	0.1565	-0.0079	0.0110	-0.0214	-0.0675
2006-2009	0.0010	0.0100	0.1287	0.0078 *	0.0216 ***	0.0172 *	0.8028	-0.1990	0.0138	-0.0763 ***	-0.1100 ***	-0.0801
2007-2009	-0.0047	0.0949	0.0278	0.0058	0.0284 ***	0.0180 *	2.9377 **	-0.7101	0.0442 *	-0.1747 ***	-0.1329 ***	-0.1678 ***

Notes: Reports the relationship between analyst forecast errors (scaled by PPS_{t-1}) and CSR performance using OLS regressions that include dummies for industry, year, missing ADV_{t-1} , missing RND_{t-1} , missing $IndADV_{t-1}$ and missing $IndRND_{t-1}$, which were suppressed for ease of presentation.
 ***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

Appendix VIII

TABLE 12: FORECAST ERRORS BY PERIOD, SCALED BY TOTAL ASSETS (T-1)

Period	CSR_{t-1}	$Q_{t-1} * CSR_{t-1}$	Q_{t-1}	Gov_{t-1}	ADV_{t-1}	RND_{t-1}	$IndADV_{t-1}$	$IndRND_{t-1}$	$Q_{t-1} * Gov_{t-1}$	$Q_{t-1} * ADV_{t-1}$	$Q_{t-1} * RND_{t-1}$	Cons.
1992-1995	0.0076 *	0.0448	0.0135	-0.0041 ***	0.0008	-0.0021	-0.2797	0.0234	-0.0384 ***	-0.0092	0.0343	0.0229
1993-1996	0.0092 **	0.0676 **	-0.0894 ***	-0.0041 ***	-0.0019	-0.0011	-0.1164	-0.3336	-0.0124	-0.0030	0.0780 ***	0.0084
1994-1997	0.0122 ***	0.0649 **	-0.1210 ***	-0.0044 ***	-0.0039	-0.0012	-0.0419	0.0740	-0.0070	-0.0016	0.0821 ***	0.0040
1995-1998	0.0165 ***	-0.0164	-0.0544	-0.0050 **	0.0022	-0.0328 ***	-0.2612	-0.6776	-0.0019	-0.0069	0.1162 ***	0.0333
1996-1999	0.0157 ***	-0.0710 **	-0.0250	-0.0031	0.0016	-0.0420 ***	-0.0948	0.0519	0.0039	-0.0247	0.1408 ***	0.0248
1997-2000	0.0090 *	-0.1404 ***	0.0501	-0.0040 **	-0.0013	-0.0478 ***	0.4766	-1.4656 ***	-0.0066	-0.02890 *	0.1432 ***	0.0118
1998-2001	0.0041	-0.1090 ***	0.0327	-0.0029	0.0023	-0.0446 ***	0.0857	-1.1473 ***	-0.0094	-0.0122	0.1245 ***	0.0447
1999-2002	0.0036	-0.0919 ***	0.0042	-0.0024	-0.0013	-0.0127 ***	0.2393	-0.7261 ***	-0.0044	0.0075	0.0824 ***	0.0014
2000-2003	0.0025	0.0036	0.0342 *	-0.0016	0.0005	-0.0004	0.1656	-0.0443	-0.0122 **	0.0078	0.0065	-0.0067
2001-2004	0.0041	0.0018	0.0549 ***	-0.0020	0.0013	-0.0005	0.1735	0.0377	-0.0145 **	-0.0046	0.0022	-0.0033
2002-2005	0.0054 *	-0.0070	0.0729 ***	-0.0006	0.0016	-0.0016	0.2343	-0.0586	-0.0123 *	-0.0078 *	0.0022	-0.0128
2003-2006	0.0069 *	-0.0071	0.0454 *	0.0012	-0.0024 *	-0.0016	-0.2019	0.0017	-0.0035	0.0024	-0.0004	0.0067
2004-2007	0.0042	-0.0382	0.0515	-0.0008	-0.0025 *	-0.0029	-0.0266	0.0996	0.0005	0.0047	0.0040	0.0060
2005-2008	0.0075	-0.0022	0.0202	0.0002	-0.0029 *	-0.0017	0.0001	0.0473	-0.0007	0.0081	0.0030	-0.0041
2006-2009	0.0086	-0.0125	0.0141	-0.0014	0.0021	0.0035	0.1500	-0.1078	0.0058	0.0012	-0.0144	-0.0078

Notes: Reports the relationship between analyst forecast errors (scaled by $Assets_{t-1}$) and CSR performance using OLS regressions that include dummies for industry, year, missing ADV_{t-1} , missing RND_{t-1} , missing $IndADV_{t-1}$ and missing $IndRND_{t-1}$, which were suppressed for ease of presentation.

***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

Appendix IX

TABLE 13: EARNINGS ANNOUNCEMENT SURPRISE BY PERIOD FOR EVENT WINDOW -5:+1

<i>Period</i>	CSR_{t-1}	$Q_{t-1} * CSR_{t-1}$	Q_{t-1}	Gov_{t-1}	ADV_{t-1}	RND_{t-1}	$IndADV_{t-1}$	$IndRND_{t-1}$	$Q_{t-1} * E_{t-1}$	$Q_{t-1} * ADV_{t-1}$	$Q_{t-1} * RND_{t-1}$	<i>Cons.</i>
1992-1995	0.0050	-0.0229	-0.0319	-0.0015	0.0036	0.0027	0.0828	-0.6694	0.0155	0.0237	-0.0194	-0.0079
1993-1996	0.0030	-0.0521 **	0.0027	-0.0015	0.0023	0.0041	0.2922	0.0219	-0.0030	0.0303 *	-0.0027	-0.0301
1994-1997	-0.0010	-0.0790 ***	-0.0198	-0.0014	-0.0033	0.0036	0.4066 *	0.1153	0.0164 *	0.0601 ***	-0.0131	-0.0289
1995-1998	0.0031	0.0381	-0.0264	0.0001	0.0001	0.0086 ***	0.4621 *	-0.0715	0.0038	0.0086	-0.0047	-0.0287
1996-1999	0.0020	0.0017	-0.0076	0.0011	-0.0009	0.0063 *	-0.0448	-0.6463	-0.0037	0.0218	-0.0005	0.0220
1997-2000	0.0017	0.0315	-0.0047	0.0016	-0.0079	0.0017	0.0431	0.1057	-0.0032	0.0292 *	-0.0099	-0.0356
1998-2001	0.0026	-0.0136	0.0451	-0.0005	-0.0018	-0.0020	0.1643	0.3474	-0.0110	0.0048	-0.0060	-0.0076
1999-2002	0.0047	-0.0625 **	0.0633 *	-0.0028	0.0045	0.0027	0.2305	-0.1382	-0.0126	0.0004	-0.0033	-0.0254
2000-2003	0.0068	-0.0676 **	0.0561 *	-0.0022	0.0046	-0.0019	0.4812	0.0241	-0.0061	-0.0127	0.0006	-0.0382
2001-2004	0.0064	-0.0667 **	0.0257	-0.0015	0.0044	0.0004	0.0663	0.1975	0.0045	-0.0019	-0.0008	-0.0424
2002-2005	0.0062	-0.0154	-0.0001	-0.0003	0.0000	0.0034	-0.0940	0.1778	0.0007	0.0045	-0.0022	0.0175
2003-2006	0.0046	0.0291	-0.0443	0.0007	-0.0014	-0.0024	-0.6614	0.1573	0.0045	0.0053	0.0108	0.0385
2004-2007	-0.0016	-0.0108	0.0295	0.0005	-0.0016	-0.0024	-0.0417	0.0387	-0.0053	0.0034	0.0069	0.0170
2005-2008	0.0003	-0.0266	0.0329	0.0026	-0.0024	-0.0022	0.6017 *	-0.0908	0.0027	0.0021	0.0100	-0.0101
2006-2009	-0.0003	-0.0476	0.0424	0.0027	-0.0019	-0.0026	0.4015	-0.2040	0.0070	0.0098	-0.0013	0.0051
2007-2010	0.0027	-0.0532	0.0461	0.0028	-0.0009	-0.0002	-0.2209	0.0129	0.0021	0.0208 *	-0.0098	-0.0141

Notes: Reports the relationship between earnings announcement returns (windows ranging -1, -3, -5, -10 and -20 to +1 trading days around quarterly earnings announcement dates), and CSR performance using OLS regression of the announcement returns in window -5:+1. We use returns in excess of the Fama-French (1993) three factors and Carhart's (1997) momentum factor over the relevant time window, using betas estimated from 21 to 250 trading days prior to the earnings announcement with a minimum of 3 trading days' data required for parameter estimation. The regression also includes dummies for industry, year-quarter, missing ADV_{t-1} , missing RND_{t-1} , missing $IndADV_{t-1}$ and missing $IndRND_{t-1}$, which were suppressed for ease of presentation. An event is Post05 if the earnings announcement date occurred during years 2005-onwards. Standard errors are reported in parentheses.

***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

Appendix X

TABLE 14: TOBIN'S Q BY PERIOD

<i>Period</i>	<i>CSR_{t-1}</i>	<i>Log(Assets)_t</i>	<i>Log(Age)_t</i>	<i>ROA_t</i>	<i>Leverage_t</i>	<i>Capex_t</i>	<i>RND_t</i>	<i>ADV_t</i>	<i>Ind-Leverage_t</i>	<i>Ind Capex_t</i>	<i>Industry RND_t</i>	<i>Industry ADV_t</i>	<i>E_{t-1}</i>	<i>Cons</i>
1992-1995	0.0150	-0.0257 *	-0.0117	0.1976 ***	-0.0174 ***	-0.0071	0.0065	0.0017	0.2202	-0.3166	1.9318	0.4353	-0.0177 ***	-0.0546
1993-1996	0.0161	-0.0201	0.0186	0.2088 ***	-0.0214 ***	-0.0188 ***	0.0247 **	0.0048	0.1425	0.6712	-2.1217	-0.3124	-0.0099 **	-0.1885
1994-1997	0.0144	-0.0171	0.0294	0.2020 ***	-0.0226 ***	-0.0131 **	0.0241 **	0.0112	0.1331	-0.2263	-1.2274	-0.8257	-0.0034	-0.1396
1995-1998	0.0321 **	0.0000	0.0326	0.2264 ***	-0.0364 ***	-0.01213 **	0.0270 **	0.0053	0.0007	0.1826	0.4299	-1.0450	-0.0070	-0.2544
1996-1999	0.0313 *	0.0361 *	0.0152	0.2264 ***	-0.0524 ***	-0.0087	0.0380 ***	0.0161	0.5811	1.1159	0.0692	-1.0938	0.0001	-0.601156 *
1997-2000	0.0422 **	0.0222	-0.077028 **	0.2323 ***	-0.0446 ***	-0.0028	0.0434 ***	0.0116	0.1169	1.0803	1.1079	1.5944	-0.0050	-0.4493
1998-2001	0.0536 ***	0.0076	-0.087499 **	0.2160 ***	-0.0552 ***	0.0040	0.0508 ***	0.0208	0.7565	-0.1872	0.7259	-0.5707	-0.0115	-0.2360
1999-2002	0.0488 **	-0.0468 *	-0.1012 ***	0.1595 ***	-0.0403 ***	0.0041	0.0573 ***	0.0568 ***	0.2431	-2.3000 *	0.9657 **	0.2731	-0.0220 ***	0.1700
2000-2003	0.0294	-0.0955 ***	-0.0791 ***	0.1364 ***	-0.0161 ***	0.0158 *	0.0524 ***	0.0293 *	0.0679	-0.9732	0.6409	-0.4787	-0.0300 ***	0.4216 **
2001-2004	0.0103	-0.0957 ***	-0.0758 ***	0.1325 ***	-0.0172 ***	-0.0008	0.0384 ***	0.0265 ***	-0.0563	-1.9010	0.8048	0.2900	-0.0203 ***	0.4686 **
2002-2005	0.0047	-0.0814 ***	-0.1055 ***	0.1469 ***	-0.0147 ***	-0.0012	0.0258 ***	0.0104 *	0.2431	-1.6359	0.8969 *	2.9634	-0.0183 ***	0.2978
2003-2006	0.0125	-0.0798 ***	-0.1091 ***	0.1710 ***	-0.0146 ***	-0.0018	0.0227 ***	0.0034	0.1259	1.1331	1.1907 **	1.5938	-0.0198 ***	0.2436
2004-2007	0.0693 ***	-0.0665 ***	-0.0694 ***	0.1404 ***	-0.0183 ***	-0.0017	0.0167 ***	0.0053	-0.5323 *	1.5931	0.2114	0.5304	-0.0130 **	0.3480 **
2005-2008	0.0935 ***	-0.0535 ***	-0.0613 **	0.1040 ***	-0.0157 ***	0.0002	0.0176 ***	0.0008	-0.3573	0.6017	-0.3261	1.1455	-0.0136 **	0.3210 *
2006-2009	0.1006 ***	-0.0562 ***	-0.050087 **	0.0848 ***	-0.0162 ***	-0.0014	0.0277 ***	-0.0060	-0.1767	-0.0717	0.0577	1.6273	-0.0168 ***	0.2993 *

Notes: Reports the relationship between Tobin's q and CSR performance using OLS regressions that include dummies for The regression also includes dummies for industry, year-quarter, missing ADV_t , missing RND_{t-1} , missing $IndADV_{t-1}$ and missing $IndRND_{t-1}$, which were suppressed for ease of presentation.

***, ** and * are significant at the 1, 5, and 10 percent level, respectively.

Appendix XI

TABLE 15: ROA BY PERIOD

<i>Period</i>	<i>CSR_{t-1}</i>	<i>Log(BME)_{t-1}</i>	<i>Log(Assets)_t</i>	<i>Log(Age)_t</i>	<i>Capex_t</i>	<i>RND_t</i>	<i>ADV_t</i>	<i>IndustryCapex_t</i>	<i>IndADV_t</i>	<i>IndRND_t</i>	<i>Gov_t</i>	<i>Cons</i>
1992-1995	0.0550	-0.4979 ***	-0.0012	-0.0618	0.0967 ***	0.0051	0.0103	-0.1848	-1.4185	-0.8808	0.02152 *	-0.1571
1993-1996	0.0497	-0.4812 ***	0.0073	-0.1475 **	0.0644 ***	-0.0195	0.0168	-0.8892	-0.5721	1.1071	0.0025	0.1017
1994-1997	0.0445	-0.5011 ***	0.0256	-0.1195 **	0.0586 ***	-0.0254	0.0072	0.0518	0.1686	-2.3456	-0.0006	-0.2024
1995-1998	0.0178	-0.4847 ***	-0.0008	-0.0789 *	0.0406 ***	-0.0155	0.0078	0.3672	-2.0466	0.1715	-0.016037 *	-0.0338
1996-1999	0.0139	-0.4794 ***	-0.0354	-0.0495	0.0354 ***	-0.0110	0.0246	-1.0683	-2.4716	0.4740	-0.0135	0.1123
1997-2000	0.0085	-0.3614 ***	-0.0371	0.0083	0.0340 ***	-0.0225	0.0258	-1.5321	-1.3898	1.2380	-0.0043	-0.1085
1998-2001	0.0117	-0.3207 ***	-0.0433	-0.0447	0.0314 ***	-0.0218	0.0086	-0.1199	1.9007	0.2018	-0.0048	-0.2835
1999-2002	0.0113	-0.2827 ***	-0.0279	-0.0241	0.0280 ***	-0.0475 **	0.0196	-0.2947	1.6746	0.5191	-0.0041	-0.3051
2000-2003	0.0273	-0.2755 ***	-0.0080	-0.0248	0.0364 **	-0.0666 ***	0.0409	0.0569	1.7795	0.6732	-0.0069	-0.3577
2001-2004	0.0579 *	-0.3299 ***	0.0339	-0.0337	0.0122 ***	-0.0739 ***	0.0242	0.6450	1.4180	1.2558	-0.0103	-0.466595 *
2002-2005	0.0743 **	-0.3800 ***	0.0557	0.0386	0.0119 ***	-0.0344 ***	0.0209 **	-0.4060	1.2995	1.915309 **	-0.0082	-0.7820 **
2003-2006	0.1014 ***	-0.4625 ***	0.0684 **	0.0805 *	0.0121 ***	-0.0215 *	0.0139 *	-2.9754	1.0500	0.0566	0.0061	-0.9062 ***
2004-2007	0.0660 *	-0.5112 ***	0.0216	0.1382 ***	0.0125 ***	-0.0156	0.0142 *	-4.690729 *	3.0255	-0.0769	0.0091	-0.8598 ***
2005-2008	0.0350	-0.5557 ***	0.0102	0.1854 ***	0.0212 ***	-0.0171	0.0074	-1.6583	-2.0059	-1.0130	0.0062	-0.8563 ***
2006-2009	0.0157	-0.3237 ***	-0.0031	0.1119 **	0.0323 ***	-0.0410 **	-0.0049	0.1994	-4.1656	-1.8365	-0.0097	-0.2512

Notes: Reports the relationship between ROA and CSR performance using OLS regressions that include dummies for industry, year, missing ADV_{t-1} , missing RND_{t-1} , missing $IndADV_{t-1}$ and missing $IndRND_{t-1}$, which were suppressed for ease of presentation.

***, ** and * are significant at the 1, 5, and 10 percent level, respectively.